

Master Thesis

Optimising Chain-of-Thought Reasoning for Cryptic Clue Solving Using LLMs as In-Context Teachers

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a thesis submitted in fulfillment of the
requirements for the degree of

MA Linguistics
(Language and AI)

Vrije Universiteit Amsterdam

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Date	June 26, 2026
Student number	2902283
Word count	16,995

Abstract

Solving cryptic crossword clues remains challenging for Large Language Models (LLMs) because the solution process requires wordplay expertise, language disambiguation, and multi-step reasoning. Despite the centrality of intermediate reasoning, previous studies in the domain have largely focused on improving answer accuracy, with limited attention to the Chain-of-Thought quality. Therefore, the present thesis aims to investigate the effectiveness of curated reasoning traces generated by a task-capable LLM as in-context demonstrations to optimise the reasoning performance of another LLM. The framework is driven by an original fine-grained LLM-as-a-judge reasoning evaluation mechanism that targets final accuracy, decomposition quality, inference soundness, and structural coherence. The results reveal a generally limited reasoning performance gain after the curated demonstrations, although this varies across wordplay categories and can be clue-dependent. Qualitative analysis suggests that the student-LLM captures procedural and structural signals from the demonstrations to a certain extent, while this remains shallow and inconsistent. Overall, the findings indicate an interplay of task complexity, data stratification, and model capability that can constrain the demonstrative effects. Consequently, these factors leave room for further exploring the effectiveness of the proposed paradigm.

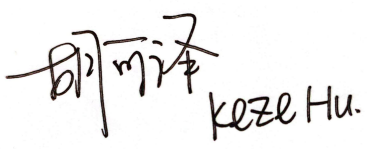
Keywords:*Large Language Models; Chain-of-Thought prompting; cryptic crossword clues; In-Context Learning; fine-grained reasoning evaluation*

Declaration of Authorship

I, Keze Hu, declare that this thesis, titled *Optimising Chain-of-Thought Reasoning for Cryptic Clue Solving Using LLMs as In-Context Teachers* and the work presented in it are my own. I confirm that:

- This work was done wholly or mainly while in candidature for a Master degree at this University.
- Where any part of this thesis has previously been submitted for a degree or any other qualification at this university or any other institution, this has been clearly stated.
- Where I have consulted the published work of others, this is always clearly attributed.
- Where I have quoted from the work of others, the source is always given. With the exception of such quotations, this thesis is entirely my own work.
- I have acknowledged all main sources of help.
- Where the thesis is based on work done by myself jointly with others, I have made clear exactly what was done by others and what I have contributed myself.

Date: June 26, 2026

Signed: 

Acknowledgments

I would like to express my deepest appreciation to my two thesis supervisors, Prof. Dr. Antske Fokkens (Vrije Universiteit Amsterdam) and Dr. Pauline van Nies (Sopra Steria). Without their invaluable expertise, patience, feedback and support, I would not have been able to complete this thesis.

Special thanks to my professors, especially Dr. Pia Sommerauer, and my classmates, including Rey Rashid and Rico Nicolass, from VU Amsterdam. Their insights are constructive to this thesis. Thanks should also go to my lovely colleagues from the Data, Science & AI departments at Sopra Steria for their practical support throughout the pipeline construction process.

Moreover, I would be remiss if I did not mention my family, especially my parents, for their unwavering love and belief in me every step of the way.

Last but certainly not least, I want to thank my wonderful friends, particularly Mary Chen, Myla Chen, Zowie He, Maggie Shi, and Steven Xu, for their moral support.

List of Figures

3.1 GHCI Row Distribution 19
3.2 Filtered Dataset Wordplay Distribution 21

4.1 Research Design Overview 24
4.2 Final Generation Prompt 32
4.3 Reasoning Trace Comparison before (top) and after (bottom) the Prompt Revision on the clue: *'Filmmaker to sensationalise hollow admission by scientist* (10)' 32

A.1 Exploratory Stage LLM Judge Prompt 75

List of Tables

3.1 GHC Example 18
3.2 GHCI Example 19
3.3 Merged Dataset Example 20
3.4 Set Labels for Experiments 22

4.1 Overview of the Evaluation dimensions. 26
4.2 Calibrated Evaluation Metric 29
4.3 Diagnostic Stage Model Selection 30

5.1 Final accuracy and average reasoning quality scores conditioned on correct answers for teacher pre-generation 36
5.2 Ordering Optimisation Experiments 36

6.1 General Model Performance 39
6.2 Final Accuracy and Number of Correct Cases per Wordplay 40
6.3 Average Decomposition Quality Scores and Number of Score Changes per Wordplay 41
6.4 Average IS Scores per Wordplay and Number of Score Changes per Wordplay 47
6.5 Average SC Scores per Wordplay and Number of Score Changes per Wordplay 50

A.1 Exploratory Experiment Basic Metric 74
A.2 Overview of Exploratory Stage Human Annotation Results 75

B.1 Judge Candidate Comparison 80
B.2 Final Calibration IAA Scores 81

Contents

Abstract i

Declaration of Authorship iii

Acknowledgments v

List of Figures vii

List of Tables ix

1 Introduction 1

1.1 Cryptic Crossword Clues 1

1.2 Research Goals and Questions 2

1.3 Relevance and Contributions 3

2 Background and Related Work 5

2.1 Large Language Models (LLMs) 5

2.2 Prompting Engineering Mechanisms 6

2.2.1 In-Context Learning 6

2.2.2 Chain-of-Thought Prompting 7

2.3 LLMs as In-Context Reasoning Teachers 7

2.3.1 Framework Utility 8

2.3.2 Application Challenges 8

2.4 Reasoning Evaluation for Optimisation 9

2.4.1 Fine-Grained Reasoning Evaluation 10

2.4.2 LLM-as-a-Judge 11

2.5 Cryptic Crossword Clues 12

2.5.1 A Challenging Task to LLMs 12

2.5.2 Towards Process-Based Supervision 13

2.6 Summary 13

2.7 Hypotheses 14

3 Data **17**
3.1 Dataset Characteristics 17
3.1.1 GHC-INDICATORS (GHCI) 18
3.2 Preprocessing 20
3.2.1 Format Transformation 20
3.2.2 Data Filtering 20
3.3 Set Splitting 21
3.4 Subsets for Experiments 22

4 Methodology **23**
4.1 Exploratory Stage 24
4.1.1 Lightweight Experiment 24
4.1.2 Preliminary Metric Design 25
4.2 Diagnostic Stage 26
4.2.1 Instrument Development 27
4.2.2 Model Selection 30
4.3 Main Implementation 31
4.3.1 Teacher Tier 31
4.3.2 Judge Tier 33
4.3.3 Student Tier 33
4.4 Evaluation 34
4.4.1 Quantitative Performance Evaluation 34
4.4.2 Qualitative Reasoning Analysis 34

5 Implementation **35**
5.1 Reasoning Pool Formation 35
5.2 Student-LLM Prompting 36

6 Evaluation Results & Analysis **39**
6.1 General Performance 39
6.2 Final Accuracy 40
6.3 Decomposition Quality 40
6.3.1 Quantitative Comparison 41
6.3.2 Qualitative Comparison 41
6.4 Inference Soundness 46
6.4.1 Quantitative Comparison 46

6.4.2 Qualitative Comparison 47
6.5 Structural Coherence 50
6.5.1 Quantitative Comparison 50
6.5.2 Qualitative Comparison 50
6.6 Summary 53

7 Discussion **55**
7.1 Sub-RQ1 55
7.1.1 Negative Utility in Task Non-Linearity 55
7.1.2 Clue-dependent effectiveness 56
7.1.3 Internal Constraints 57
7.2 Sub-RQ2 58
7.2.1 Structural Mimicry 58
7.2.2 Procedural but Non-Competent Learning 59
7.3 Limitations & Future Work 59

8 Conclusion **61**

References **63**

A Lightweight Experiment (Exploratory) **73**
A.1 Preliminary Dataset 73
A.2 Prompt and Configuration 73
A.2.1 Simplistic Modelling Prompt 73
A.2.2 Model Configuration 73
A.2.3 Reasoning Outputs 74
A.2.4 Preliminary Model Behaviour Exploration 74

B Instrument Development (Diagnostic) **77**
B.1 Initial Calibration 77
B.1.1 LLM Judge Configuration 77
B.1.2 Metric 77
B.1.3 Results 78
B.2 Middle Calibration 78
B.2.1 LLM Judge Configuration 78
B.2.2 Metric 79
B.2.3 Results 80
B.3 Final Calibration 81

B.3.1 LLM Judge Configuration 81
B.3.2 Final Metric 81
B.3.3 Results 81

Chapter 1

Introduction

The deployment of Large Language Models (LLMs) has surged in many fields. State-of-the-art LLMs have achieved improved performance on various mathematical and scientific benchmarks, such as calculation and common sense reasoning. Chain-of-Thought (CoT) prompting, a paradigm in which LLMs generate a series of intermediate reasoning steps leading to the final answer (Wei et al., 2022b), plays a pivotal role in this improvement.

However, the predominantly final-answer-based evaluation metric used in these benchmarks can lead to the reasoning capacity and quality of LLMs being overlooked (Shojaee et al., 2025). With the correct final answers, the generated reasoning traces can still be fallacious and incoherent (Lee and Hockenmaier, 2025; Barez et al., 2025). Therefore, there have been shifts in the field towards more process-based than accuracy-based supervision and optimisation, particularly for complex reasoning tasks, that emphasise the intermediate stepwise reasoning quality to mitigate hallucination and improve trace explainability (Lightman et al., 2024).

The present thesis follows this trend and investigates CoT quality optimisation for solving cryptic crossword clues, a domain that reflects complexity and necessitates high-quality reasoning. Misleading surface readings, as well as ambiguously encoded wordplay (e.g., anagrams and containers), enable cryptic clues to be 'a satisfactory intellectual challenge' (Andrews and Witteveen, 2025, p.1) to both humans and LLMs (Efrat et al., 2021).

1.1 Cryptic Crossword Clues

A concrete cryptic crossword clue is demonstrated in (1):

(1) *Country left judgeable after odd losses* (8); Answer: PORTUGAL

Here, the goal is not to directly find a country that 'left judgeable after odd losses'. Instead, it requires the decomposition and/or disambiguation of the following components in a clue:

- Definition: One or more words in the clue that can be interchangeably used with the final answer, such as 'country' in (1) with respect to the answer 'Portugal' (a country).
- (Enumeration): Number of letters required in the answer. An eight-letter word referring to a country is needed in (1), for example.
- Wordplay: chunks in the clue that require disambiguation to find an alternative spelling or expression of the same word. In (1), 'left' can be associated with 'port', the left side of a ship or aircraft as you are facing forward.

  The wordplay chunk also includes an indicator that implicitly instructs the wordplay type required for answer parsing. In (1), 'after odd losses' maps to alteration, requiring modifications to the letters at odd positions.
- Fodder: A chunk in the clue functions as a manipulation source for (part of) the final answer. When applying odd-letter removal to 'judgeable', we get 'ugal' in (1). After concatenation, 'PORTUGAL' is indeed an eight-letter named entity for a country.

The construction and inference mechanism demonstrated above thus reveals the need for wordplay expertise, world knowledge, and creativity for coherent, stepwise clue segmentation and cross-componential inference to reach the final answer (Sadallah et al., 2025).

However, without external guidance on clue deconstruction and inference structure, cryptic clue solving remains a challenge for LLMs due to their limitations in less-constrained multi-step reasoning over ambiguous language (Sadallah et al., 2025; Shojaee et al., 2025). These limitations suggest there is space for reasoning optimisation, which is insufficiently studied in the domain. This, therefore, motivates the current study to explore the effectiveness of a prompt-based guided approach to optimising reasoning in the domain of cryptic-clue solving.

1.2 Research Goals and Questions

McDonald and Emami (2024) have shown that deploying more powerful LLMs as in-context teachers improves the reasoning abilities of another model to solve complex

reasoning tasks, such as mathematical reasoning. Given this advantage, this thesis aims to explore the effectiveness of integrating curated reasoning traces by a teacher LLM as few-shot demonstrations in optimising the CoT for solving cryptic clues.

Now that the overall goal is established, the following research questions (RQs) are to be addressed in the current work:

- **Main RQ**: To what extent can reasoning traces generated by a teacher LLM, curated as in-context expert-level demonstrations, improve the reasoning performance of another LLM when solving cryptic crossword clues?
- **Sub-RQ1**: How does this prompt-based optimisation improve performance compared to standard CoT prompting across different wordplay types?
- **Sub-RQ2**: Which reasoning structures or decomposition strategies emerge in student LLMs' reasoning traces with externally curated guidance?

1.3 Relevance and Contributions

In exploring the research questions above, the present thesis contributes at the theoretical level by yielding insights into model reasoning capabilities beyond final accuracy in the domain of cryptic-clue solving.

Practically, this thesis developed a rigorous prompting paradigm, pivoted by a fine-grained LLM-as-a-judge mechanism to evaluate and select quality reasoning traces to solve cryptic clues as demonstrative candidates for student model optimisation.

Further, the current work extends the evaluation domains of the prompting paradigm by (McDonald and Emami, 2024), showing its potentially limited efficacy in linguistically non-linear and heterogeneous domains.

The thesis is structured as follows. Chapter 2 establishes the theoretical basis of prompt engineering and reasoning optimisation and reviews existing studies related to cryptic clues, contributing to the formulation of research questions and hypotheses. This is followed by presenting data preprocessing (Chapter 3) and framing the research design (Chapter 4. The implemented system is presented in Chapter 5, which is quantitatively and qualitatively evaluated in Chapter 6. Finally, the thesis explicitly answers the research questions, together with a critical reflection on the results and implications for future work.

Chapter 2

Background and Related Work

The present thesis aims to investigate LLM reasoning optimisation for solving cryptic crossword clues. To contextualise the investigation, this chapter aims to establish the theoretical background. First, I outline the fundamental architecture and principles of generative LLMs, followed by a discussion on relevant prompting mechanisms and their limitations. Subsequently, I examine empirical studies on reasoning evaluation. Furthermore, I review the current findings with respect to the target domain (i.e., cryptic clue solving). The chapter concludes with a synthesis of key findings and research gaps, which directly inform the formulation of the research questions and hypotheses.

2.1 Large Language Models (LLMs)

In this paper, LLMs refer to transformer-based models with large parameters. The underlying transformer architecture is a deep learning structure with multiple layers that allows LLMs to process input globally rather than word-by-word (Vaswani et al., 2017). Modern LLMs, such as GPT, often utilise the decoder-only variant of the transformer-based architecture for text generation. They take the entire token sequence as input to probabilistically predict the next token based on the preceding context (Brown et al., 2020).

During pre-training, LLMs are exposed to a vast amount of language data to learn statistical regularities and semantic relationships between words (Naveed et al., 2025; Mikolov et al., 2013). In the context of cryptic crossword clues, these learned representations facilitate the retrieval of a synonym to the *definition* or a closely related wordplay mechanism to the *indicator*.

At inference time, the internal attention mechanism simultaneously computes weights representing token associations within a sequence (Vaswani et al., 2017). However,

this process inherently lacks a sense of word order. To resolve this, positional encodings are injected to provide unique mathematical vectors (e.g., cosine functions) to represent the absolute or relative position of each token. Residually connected with the weights calculated, initial token embeddings are transformed into dynamic contextual representations with the order preserved. This is crucial for cryptic clue solving, as subtle sequential differences can hint at different wordplay and answers (Sadallah et al., 2025).

By featuring these, LLMs can be conditioned to address various NLP tasks, such as cryptic crossword clues. Since modern LLMs are largely instruction fine-tuned, they can straightforwardly generate text conditioned on the requirements (e.g., expected inference formats and constraints) provided in the user prompts passed to the model (Naveed et al., 2025). In particular, with induction heads, a specialised attention structure, LLMs can recognise patterns if given demonstrations and repeat them in the next generation (Crosbie and Shutova, 2025).

Moreover, their intended purposes can constrain their generations. This is exemplified by the GPT reasoning models, which can test with alternatives and correct mistakes internally before the final generation (OpenAI, 2026). This reasoning pathway differs from that of models designed for straightforward generation.

Empirical evidence has revealed that prompts can enhance model generation and performance (e.g., (Brown et al., 2020)). However, this enhanced performance is susceptible to prompt quality, including structure and formatting. Consequently, such vulnerability necessitates careful prompting design and optimisation, which is further discussed in the next sections.

2.2 Prompting Engineering Mechanisms

2.2.1 In-Context Learning

LLMs benefit from learning signals in prompts to address complex tasks, conceptualised as In-Context Learning (ICL). Prompting LLMs with learning signals can be achieved by one or more clear demonstrations that map a given input to a desired output. With these demonstrations, also known as *few shots* (Brown et al., 2020), LLMs learn to address a new task without updating weights and parameters. The transformer-based architecture of LLMs described in Section 2.1 is believed to support this learning capacity (Dong et al., 2024).

Brown et al. (2020) have demonstrated that prompting with demonstrations improves model performance on various NLP benchmarks (e.g., translation) and on tasks

requiring on-the-fly capabilities (e.g., word manipulation). These findings suggest that few-shot prompting is a promising optimisation approach, deserving further exploration of its application to enhance model performance in cryptic-clue solving.

However, the underlying correlation between input and output is not explicitly elicited. Its side effect is prominent in complex reasoning tasks (Dong et al., 2024), where intermediate rules or steps cannot be explicitly derived from the input. Consequently, a lack of intermediate reasoning can limit a model's ability to generalise beyond shallow pattern matching (Dong et al., 2024), thus motivating the deployment of Chain-of-Thought (CoT) prompting.

2.2.2 Chain-of-Thought Prompting

CoT prompting guides LLMs to generate intermediate reasoning before final answers, utilising their emergent ability to mimic how humans deconstruct a complex task into operational and logical steps (Wei et al., 2022b,a). With this, LLMs have demonstrated improved performance, particularly in mathematical and scientific reasoning task benchmarks (Wei et al., 2022b).

There are two main approaches to CoT prompting, either with explicit instructions only (e.g., 'Let us think step-by-step') (Kojima et al., 2023), or with additional reasoning exemplars (Wei et al., 2022b). The second approach integrates ICL learning (Brown et al., 2020) to provide learning signals to generate similar reasoning traces. However, their added value to the performance of LLMs remains unclear (Cheng et al., 2025b), leaving room for investigation in the domain of solving cryptic clues.

Few-shot CoT prompting utilises human-crafted rationales as quality reasoning demonstrations. Although integrating such rationales has been proven beneficial for model optimisation, their utility can be affected by human subjectivity and practical constraints (e.g., cost and time) (Wei et al., 2022b; Zhao et al., 2024). Moreover, transferability of certain cognitive patterns or structures in human-crafted rationales can vary across different tasks (Yao et al., 2023). These considerations motivate investigating rationales crafted by LLMs as a possibly additional source of reasoning demonstrations in CoT prompting, as will be explained in detail in Section 2.3.

2.3 LLMs as In-Context Reasoning Teachers

Utilising LLM-crafted rationales as demonstrations for another LLM during CoT prompting leverages both in-context teaching and learning (Zhao et al., 2024; Brown et al.,

2020). This is termed *Trace-of-Thought*, a prompt-based approach to model reasoning optimisation by McDonald and Emami (2024).

In this process, a stronger teacher LLM is prompted to pre-generate multi-step reasoning traces reflecting structured task construction, logical inference, and final answers. Mirroring a pedagogical process, these traces are curated and augmented with task descriptions as learning resources to guide a student LLM. Recent work has revealed both the potential and limitations of this paradigm, as discussed in the following section.

2.3.1 Framework Utility

Existing studies have shown the benefits of deploying LLMs as in-context teachers to enhance self- or another LLM's performance on complex tasks, such as medical question-answering and mathematical reasoning (Zhao et al., 2024; McDonald and Emami, 2024). This empirical evidence suggests Trace-of-Thought as a promising approach to optimise model reasoning for cryptic-clue solving.

Moreover, Trace-of-Thought offers advantages over the standard few-shot CoT prompting. Having a teacher-LLM generate reasoning exemplars is not only more cost-effective (Wei et al., 2022b), but can better resemble the internal pathways of the student-LLM (Zhao et al., 2024). Thus, utilising such a paradigm can help mitigate uncertainty in practical constraints and reasoning compatibility compared to manually crafted rationales, as discussed in Section 2.2.2.

Further, while inheriting the fundamental goals of knowledge distillation through fine-tuning (Hinton et al., 2015), this prompt-based reasoning transfer can reduce computational burdens and allow human oversight (McDonald and Emami, 2024). These add value to the utility of adapting Trace-of-Thought in the present thesis.

However, since Trace-of-Thought exploits both CoT and ICL, challenges closely related to the latter two mechanisms should not be overlooked.

2.3.2 Application Challenges

Selecting quality CoT demonstrations by the teacher-LLM is one of the central challenges involved in utilising the above framework. CoT is vulnerable to error propagation in both early (Tyen et al., 2024) and late (Zhang et al., 2026) stages. These errors tend to originate from hallucinations (i.e., factually incorrect or semantically irrelevant content) with high confidence (Cheng et al., 2025a). Meanwhile, LLMs can be self-inconsistent due to information drift in long contexts (Li et al., 2024).

This quality concern is compounded by empirical evidence that the actual reasoning for a model's final answer may be under- or misrepresented in the generated trace (Turpin et al., 2023). This includes 1) illogical shortcuts (i.e., directly concluding from the premise without passing and generating the key reasoning step); 2) silent correction (i.e., internally correcting a mistake without reflecting it in the explanation); and 3) post-hoc rationalisation (i.e., sycophantising the rationale towards explicit answers/subtle biases introduced in the prompts) (Barez et al., 2025; Turpin et al., 2023). Consequently, these issues can downgrade the demonstrative effects (Dong et al., 2024), which requires a careful design of the teacher trace evaluation and selection mechanism.

In addition, determining an appropriate demonstration context for the student model can be challenging. In terms of quantity, too few or too long examples may not yield an optimal demonstration effect, whereas a sheer increase can risk diminishing returns and overfitting (Dong et al., 2024). Meanwhile, LLMs can show differences in their sensitivity to the ordering of the exemplars (Lu et al., 2022). This, therefore, shows the importance of considering model-dependent factors in the formatting of demonstrations (Ratner et al., 2023).

The empirical studies reviewed above support the hypothesis that the Trace-of-Thought is an adoptable and transparent paradigm for the present thesis. However, its scaffolding and execution require careful consideration. Among others, the need for reasoning quality in teacher-LLM demonstration curation requires systematic, process-orientated supervision rather than mere final-accuracy supervision (Lightman et al., 2024).

2.4 Reasoning Evaluation for Optimisation

Reasoning evaluation is instrumental to the larger goal of reasoning optimisation, particularly for complex tasks involving multi-step reasoning (Lightman et al., 2024). Its implementation relies on a careful design of the evaluation paradigm.

Existing mathematical and coding-benchmarked evaluation metrics demonstrate their limitations in facilitating a nuanced understanding of model reasoning, primarily due to their tendency to focus on final-answer accuracy (Shojaee et al., 2025).

However, there are metrics aimed at evaluating stepwise reasoning, such as ROSCOE (Golovneva et al., 2022). It serves as a reference-free, unsupervised metric for evaluating step-by-step reasoning chains through semantic alignment, semantic similarity,

language coherence, and logical inference.

Although such metrics offer scalable evaluation, they are still probabilistic and may even produce misleading assessments of model performance (Zhou et al., 2025). Moreover, they are less dynamic and less granular for customised criteria than human evaluations (Kim et al., 2024; Papineni et al., 2002).

These limitations motivate adopting a fine-grained approach that examines reasoning quality beyond holistic metrics (i.e., evaluating overall performance), as discussed in the following section.

2.4.1 Fine-Grained Reasoning Evaluation

Mirroring the human evaluation process, a fine-grained evaluation paradigm decomposes trace quality into predefined capability dimensions (Ye et al., 2024). Empirical metrics or schemes have commonly adopted, inter alia, the following four interrelated criteria to conceptualise a quality reasoning trace (Lee and Hockenmaier, 2025): factuality, validity, coherence, and utility.

To begin with, **factuality** evaluates whether the reasoning is conditioned on the query and on the broader real-world factual information. For example, in automated text summarisation, the groundedness of a reasoning trace can be measured in binary versus the facts and context presented in the source (Lee et al., 2025). As a result, this dimension targets the quality threat of hallucination (Cheng et al., 2025a) discussed in 2.3. However, evaluating the factual knowledge that is implicit in the query can be less feasible (Lee and Hockenmaier, 2025), which requires additional (atomic) knowledge units (Prasad et al., 2023; Hwang et al., 2025)

Then, **validity** and **coherence** measure the logical soundness and self/source consistency of an LLM's reasoning, respectively. Existing studies have adopted a global or a local approach to address these two dimensions. The former benchmarks against the final answer in the trace to identify reasoning gaps, such as incompleteness and misalignment in the key-steps (Golovneva et al., 2022). Although important, global validation can mask intra-reasoning errors and information drifts (Rani Samineni et al., 2025; Li et al., 2024).

This limitation motivates a stepwise verification of logical correctness, entailment, or completeness given the premise (Lightman et al., 2024; Tyagi et al., 2024). Consequently, both global and local validity can be used to help detect the propagation of errors, hallucinations, and invalid shortcuts outlined in Section 2.3.

Further, **utility** focuses on the necessity of a reasoning step to the final answer in the trace, which helps reveal redundancy and repetition (Prasad et al., 2023). Hence,

recalling curating demonstrations for few-shot CoT prompting, insights into this dimension contribute to goal-orientated and functionally efficient candidate reasoning traces for demonstrations (McDonald and Emami, 2024; Dong et al., 2024).

These dimensions establish a fundamental basis for a more nuanced evaluation of LLMs' reasoning traces in the present thesis. However, these dimensions remain generic because the importance, conceptualisation, and feasibility of these dimensions vary across domains. For example, determining whether a reasoning step is valid differs in mathematical reasoning and more open-ended tasks. This therefore suggests a careful design of criteria that is targeted at the task in question.

Beyond conceptualising what constitutes a quality reasoning trace, the appropriate selection of evaluators facilitates a principled evaluation of reasoning. Existing evaluation paradigms predominantly involve humans and/or static-based models for fine-grained reasoning evaluation. Although human evaluations offer more reliable and nuanced insights than static-based evaluators (Kim et al., 2024), they are less cost-effective in downstream applications (Lee and Hockenmaier, 2025).

2.4.2 LLM-as-a-Judge

Balancing both granularity and scalability, Zheng et al. (2023) propose the 'LLM-as-a-judge' methodology. In this methodology, a judge LLM is tuned (e.g., fine-tuning or reinforcement learning) or strategically prompted (e.g., rule augmentation) to be able to score, select and rank the given input (Li et al., 2025).

Existing studies have shown the effectiveness and alignment with humans in deploying LLM-as-a-Judge to evaluate the quality of other LLMs with customised metrics, such as text summarisation (Song et al., 2024), free-form question-answering (Badshah and Sajjad, 2025) and rewarding stepwise reasoning quality (Lightman et al., 2024). These findings underpin its deployability to evaluate reasoning traces to solve cryptic clues.

However, limitations and challenges of integrating this methodology for fine-grained evaluation should not be overlooked. This primarily concerns the reliability and fairness of the evaluation, as LLMs can be biased towards certain types of text, style, and surface features (Li et al., 2025). This is compounded by their vulnerability to and manipulation of biased or even harmful content encoded in the prompt and training data (Bender et al., 2021).

Moreover, a judge-LLM can be less self-consistent, particularly in using a large scale for pointwise evaluation, reducing evaluation validity and reliability (Wang et al., 2025). In addition, LLMs can vary in their domain expertise, generalisability, fairness,

and adaptability (Huang et al., 2025; Li et al., 2025), underscoring the need for careful selection under task constraints.

Based on the advantages of fine-grained evaluation and the capabilities of LLMs as judges, they can jointly function as an intermediate mechanism to evaluate and select quality reasoning traces from a teacher-LLM's output as few-shot demonstrations for student prompting. However, concerns about model bias are prominent, implying the need for human oversight despite process automation.

Furthermore, the challenges revealed above about the lack of domain-specific evaluation criteria and the divergent quality of LLM-as-a-judge across domains underscore careful, domain-based considerations of what to evaluate, how to operationalise, and who to judge. Therefore, the following part reviews studies specific to cryptic crossword clue solving, a controlled linguistic task environment in the present thesis for reasoning optimisation.

2.5 Cryptic Crossword Clues

As demonstrated in Section 1.1, solving cryptic crossword clues embeds structured and logical reasoning, including segmenting components and disambiguating wordplay. This characteristic enables them to be a complex environment for testing and reasoning optimisation (Shojaee et al., 2025; Andrews and Witteveen, 2025).

2.5.1 A Challenging Task to LLMs

Both Efrat et al. (2021) and Rozner et al. (2021) have shown that LLMs achieve low final accuracy when solving cryptic crossword clues, particularly compared to human expert solvers.

There have been attempts to explain why LLMs struggle to solve cryptic crossword clues. A primary challenge lies in LLMs' frequent failure to adhere to *enumeration* in the clue (i.e., the number of letters required), a fundamental constraint of the final answer. Another factor can be attributed to LLMs' insufficiency to deconstruct the task into necessary atomic units by themselves (Sadallah et al., 2025).

Despite the generally limited competence, LLMs' performance can vary in addressing key aspects of cryptic clue solving. Sadallah et al. (2025) have revealed that LLMs perform better in surface-component decomposition (e.g., identifying definitions and generating synonyms) than ambiguity inference (e.g., identifying wordplay

types), which can be linked to their pre-training objectives and internal architectures introduced in 2.1. In addition, LLMs have been shown to be more sensitive to wordplay involving substring identification and segmentation, such as (alternating) anagrams, hidden words, and containers (Sadallah et al., 2025).

To address these challenges, empirical studies have suggested engineering prompting techniques, such as providing additional task-orientated instructions (e.g., wordplay definitions and frequent indicators) and deploying CoT (**?**), to enhance final accuracy.

2.5.2 Towards Process-Based Supervision

The studies reviewed above focus on LLMs' performance in solving cryptic clues only from the perspective of final accuracy. To date, in the domain of cryptic clue solving, there is limited process-based supervision and optimisation of LLMs' intermediate reasoning capabilities for stepwise construction of the final solution. However, the quality of stepwise reasoning is indispensable in solving cryptic clues (Efrat et al., 2021).

Andrews and Witteveen (2025) have attempted a Python-based codification to optimise step correctness. In this process, a prompted LLM translates pre-generated *wordplay* and *definition* candidates into reusable Python code composed of explicit domain-specific assertions (e.g., `assert is_synonym()`). Subsequently, an automated abstract syntax tree verifier executes this code to generate error messages with feedback used for an iterative refinement. Although this pipeline offers a seemingly stepwise validation, it remains static and accuracy-based in nature. Each key aspect of solving a cryptic clue is validated through the final proof rather than the intermediate reasoning process leading to the *definition* and *wordplay* derivations.

Consequently, existing efforts in the prior work that move beyond accuracy to reasoning quality remain scarce, leaving a research gap for the current work to address.

2.6 Summary

The present thesis treats the solving of cryptic clues as a controlled and complex domain (McDonald and Emami, 2024; Lee and Hockenmaier, 2025). Existing studies in the domain have revealed LLMs' low accuracy in solving cryptic clues, which can be attributed to their limitations in enumeration adherence, implicit inference, and wordplay disambiguation (Badshah and Sajjad, 2025; Sadallah et al., 2025). However, these

insights are predominantly accuracy-based, leaving the pivotal intermediate reasoning process for answer construction underexplored.

To address this research gap, the present work explores systematic reasoning optimisation to enhance model performance in solving cryptic clues. The literature reviewed in this chapter has demonstrated that leveraging few-shot CoT prompting can facilitate LLMs' in-context learning to solve complex tasks (Brown et al., 2020; Wei et al., 2022b,a). However, human-made examples can be expensive and not compatible with the internal pathways of LLMs (Yao et al., 2023).

Given this notion, the deployment of LLMs as in-context teachers offers a promising source for demonstrations (Zhao et al., 2024). This paradigm leverages curated synthetic reasoning traces by a stronger LLM as reusable demonstrations to guide other LLMs, enabling a more interpretable transfer of procedural reasoning behaviours than fine-tuning (McDonald and Emami, 2024).

However, whether this paradigm is effective in enhancing model performance in cryptic clue-solving is yet to be explored, motivating the main research question presented in 1.2. Meanwhile, given the divergence in LLMs' sensitivity to different wordplay in cryptic clues (Sadallah et al., 2025), it is interesting to explore whether this paradigm is effective across all wordplay types (Sub-RQ1) and what reasoning patterns can be learned by the student-LLM (Sub-RQ2).

Implementing this paradigm can present challenges due to issues related to CoT quality (Cheng et al., 2025a; Turpin et al., 2023) and demonstration prompting (Dong et al., 2024). A fine-grained LLM-as-a-judge framework (Lee and Hockenmaier, 2025; Zheng et al., 2023) can be instrumental, aiming to provide an automated granular reasoning trace scoring and selection. But these benefits rely on domain-specific evaluation criteria, consistent operationalisation, and human oversight. Thus, these findings provide practical implications for the research design described in Chapter 4.

2.7 Hypotheses

Based on the literature reviewed above, the following hypotheses are formed for the research questions presented in Section 1.2:

1. The optimised student-LLM achieves higher cryptic clue-solving accuracy and higher reasoning-quality scores than the baseline student-LLM in general.
   This hypothesis is based on the claimed efficacy of Trace-of-Thought in improving the student-LLM's problem-solving capacitities by (McDonald and Emami, 2024).

2. The optimisation effect varies across cryptic clue types, with larger improvements expected for wordplay requiring letter manipulations.
   This speculation is motivated by the better identification of the LLMs' wordplay categories involving letter assembly (Sadallah et al., 2025). Reasoning optimisation can add value to this.

3. Procedural reasoning strategies for clue analysis, rather than domain-specific and world knowledge, will be transferred to the student-LLM.
   This considers the limitations of few-shot learning in constraining outputs rather than updating internal parameters (Brown et al., 2020)

Chapter 3

Data

This chapter details the dataset used to validate the hypotheses established at the end of the previous chapter. A derived dataset collected by George Ho [1] is used, which is a publicly available collection of cryptic crossword clues. I will describe the characteristics of the data, followed by the preprocessing procedures and set splitting. Finally, there is an overview of the (sub-)sets used for the experiments.

3.1 Dataset Characteristics

This cryptic clue dataset, referred to as GHC below, encompasses more than half a million English cryptic crossword clues scraped from various web-based resources, predominantly American and British newspapers (e.g., *The New Yorker* and *The Guardian*). This means that most clues reflect American and British English and might require corresponding cultural knowledge to solve them.

GHC is formatted in a fixed pattern, with each row representing not only clue-orientated data (e.g., clue texts and publication date) but also annotations of *answers* and *definitions* (a word or a phrase in a clue that can be interchangeable with the final answer). Table 3.1 demonstrates what the data format looks like for a specific clue.

Moreover, it comprises two derived datasets: INDICATORS and CHARADES. This thesis adapts the former as the primary basis for dataset creation, as further described below.

[1] https://cryptics.georgeho.org/

Row_id	Clue	Answer	Definition	Nr	Date	Puzzle Name	URLSource
317829	*Old man greedy now and then (4)*	AGED	Old	26d	2020-08-21	Daily Telegraph 29448	Link bigdave44

Table 3.1: GHC Example

3.1.1 GHC-INDICATORS (GHCI)

GHCI [2] derives the same text and language characteristics of the parent dataset (GHC) but contains extra annotations on *indicator* and *wordplay* type for each clue.

Formats and Statistics

Unlike the parent set, GHCI is indicator-central, as the name suggests. It contains 15,735 rows categorised into eight wordplay types, including the following eight:

- **Anagram**: Rearranging letters in a word or phrase to form a different word or phrase.
- **Insertion**: Embedding one word or letter sequence directly inside another.
- **Container**: Placing a word or letter sequence inside or outside another one.
- **Reversal**: Taking a specific letter sequence and spelling it backwards.
- **Hidden**: Finding a letter sequence that appears consecutively within part of the clue text.
- **Deletion**: Removing a letter sequence from a word or phrase.
- **Homophone**: Finding a word that shares a similar pronunciation.
- **Alternation**: Selecting letters according to a pattern, such as at odd positions.

As shown in Figure 3.1, 'anagram' is the most dominant wordplay type in GHCI (7121 rows), while 'alternation' is the rarest class (244 rows).

Within each wordplay category, the clues that share the same indicators are grouped together, as shown in Table 3.2. These clues are presented in their GHC ids rather than in their textual forms.

[2] https://cryptics.georgeho.org/data/indicators

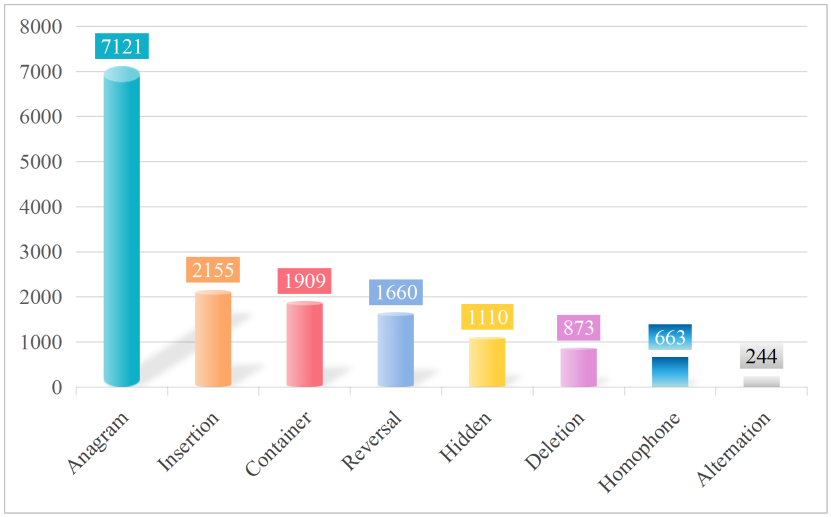


Figure 3.1: GHCI Row Distribution

Row_id	Wordplay	Indicator	Clue_rowids
34	Alternation	*Every now and then*	352813, 404850, 449145, 479724, 645784, 646340, 662594, 663868

Table 3.2: GHCI Example

Annotations

As outlined above, GHCI contains important information about wordplay categories and indicators for clues in GHC, which were available in the scraped resources. Although the dataset creator has reviewed a sample of annotations, there can still be errors and empty fields in GHCI. For example, my colleague and I noticed that some indicators cannot be found in the mapped clues when we explore the data. This thus necessitates a filtering process before final usage.

Ethical concerns

There are no serious ethical concerns about GHCI. All clues have been published, and the resources have been documented in GHC. In addition, the clues were curated for broad public audiences, which means that there was no intentional physical or mental harm. However, as George Ho admits, the fodder (i.e., chunks in a clue that must be rearranged) and logic involved in disambiguating wordplay might be offensive.

3.2 Preprocessing

With the characteristics described above in mind, I preprocessed GHCI in collaboration with another colleague. This includes format transformation and set splitting.

3.2.1 Format Transformation

As described above, GHCI features a one-to-many relationship between an indicator and its associated clues, necessitating a reformatting for subsequent experiments. To achieve this, each indicator row in GHCI was duplicated according to the number of clue IDs in the last field. As a result of this duplication, each row contained only one pointed-to clue_rowid.

Meanwhile, to facilitate dataset readability and later analysis, we merged information and annotations from GHCI and its parent dataset. Here, the clue IDs in each row were used to identify the corresponding clue rows in the parent dataset, followed by extractions of clue texts, definitions, and answers. Consequently, the original GHCI dataset was transformed into a structured six-column dataset with 92,787 rows, as demonstrated in Table 3.3.

Row_id	**Clue**	**Definition**	**Answer**	**Wordplay**	**Indicator**
317829	*Old man greedy now and then (4)*	Old	AGED	alternation	now and then

Table 3.3: Merged Dataset Example

3.2.2 Data Filtering

Based on our preliminary exploration regarding indicator annotation errors, as well as practices taken by Efrat et al. (2021), rows reflecting the following patterns were automatically removed from the merged dataset:

- Rows with empty clue and/or answer columns;
- The annotated indicator is not derived from the clue;
- Clues with enumeration (i.e., letter grid pointers) suggesting multi-word (e.g., '(2,5)') and/or non-fixed (e.g., '(2-5)') answers.

However, rows with only the *definition* field empty were preserved. The mere absence of this annotation was considered not to significantly cause negative outcomes in

the generation and evaluation of model reasoning, since it can be easily reconstructed from the clue text itself.

As a result, the filtered dataset contains 62,918 instances, the distribution of which is demonstrated in Figure 3.2. However, this filtering does not guarantee an error-free dataset. Other errors, such as wrong *definitions* and *answers*, were not checked due to infeasibility.

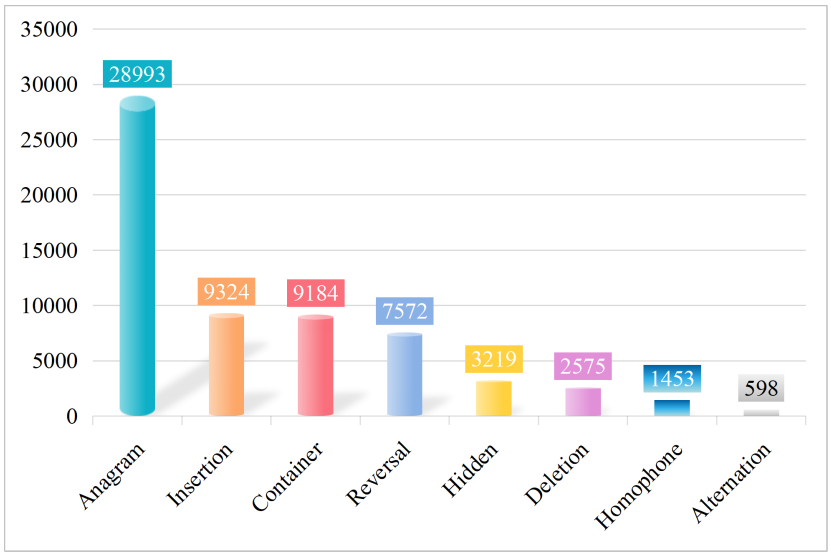


Figure 3.2: Filtered Dataset Wordplay Distribution

3.3 Set Splitting

We split the preprocessed GHCI into training, validation, and test sets. Rather than a direct ratio-based split, the test set with approximately 500 instances was first sampled. Following Sadallah et al. (2025), an equal allocation stratified sampling approach was used for an identical number of instances per wordplay type in the test set. This practice accounted for the originally imbalanced wordplay distribution, aiming to obtain sufficient samples for intra- and inter-wordplay analyses. This thus yielded a test set with 496 instances, with 62 instances per wordplay, which was held out for final evaluation.

The remaining instances were then divided into training (49,937) and validation (12,485) datasets using an 80%-20% split, without distribution normalisation. We also checked that there are instances for the smaller classes (i.e., ‘hidden’, ‘deletion’, ‘homophone’, and ‘alternation’) in both sets.

3.4 Subsets for Experiments

Given the notion that Trace-of-Thought-style reasoning optimisation prioritises quality demonstration curation and does not require substantial data compared to fine-tuning (McDonald and Emami, 2024), I further sampled subsets from the training and validation sets for my experiments.

Table 3.4 presents a complete overview of all set labels with a brief description of their sizes and functions for later reference.

Set Labels	Size	Description
`preliminary_data`	40	Randomly picked instances at the data exploration stage. This set is used for preliminary insights into model behaviour, instrument development and model selection. There are no instances from the test set.
`teacher_data`	400	Derived from the training set, this dataset reflects an equal distribution. This set is used for generating teacher reasoning trace candidates.
`optimiser_data`	200	Derived from the validation set, this dataset is used for an automatic student prompt optimisation to find an optimal ordering of trace demonstrations.
`test_data`	496	As described in the section above, this set is held out for main evaluation and analysis.

Table 3.4: Set Labels for Experiments

The complete research design, including how each subset of the data is utilised, will be explicated in the following chapter.

Chapter 4

Methodology

The core task of the present thesis is to explore the effectiveness of synthetic reasoning traces, curated as few-shot guidance, in optimising another LLM's intermediate answer-construction process for its performance in solving cryptic crossword clues.

To facilitate such a model-centred pedagogical scenario (Zhao et al., 2024; Brown et al., 2020), the *Trace-of-Thought* prompting strategy by McDonald and Emami (2024) was adapted as the primary prompting paradigm. In their paradigm, reasoning traces generated by a capable LLM with more parameters are used as demonstrations for few-shot CoT prompting with a smaller LLM as the student. However, departing from this paradigm, I redefined the pedagogical roles based on differences in models' capabilities to solve a cryptic clue, with both correct answers and structured intermediate reasoning traces, rather than on the number of parameters.

In addition, the present work contributes to an approach to mitigating the negative effects of low-quality reasoning demonstrations beyond the original paradigm. I designed a fine-grained, rubric-like reasoning evaluation scheme and embedded it with the LLM-as-a-Judge framework. Consequently, as visualised in Figure 4.1, these modifications introduced a three-tier prompting paradigm. Section 4.3 further describes this.

For the presented paradigm to work, the quality of the models' multi-step reasoning traces for solving cryptic clues is important. However, as discussed earlier, this is largely underexplored in the domain (Efrat et al., 2021; Sadallah et al., 2025; Andrews and Witteveen, 2025). Therefore, I carried out *exploratory* studies to obtain insights into models' domain-specific reasoning patterns, which facilitated the definition of evaluation criteria.

Moreover, a suitable model selection and a reliable instrument for fine-grained evaluation are integral to the paradigm. Hence, I implemented a *diagnostic* stage before

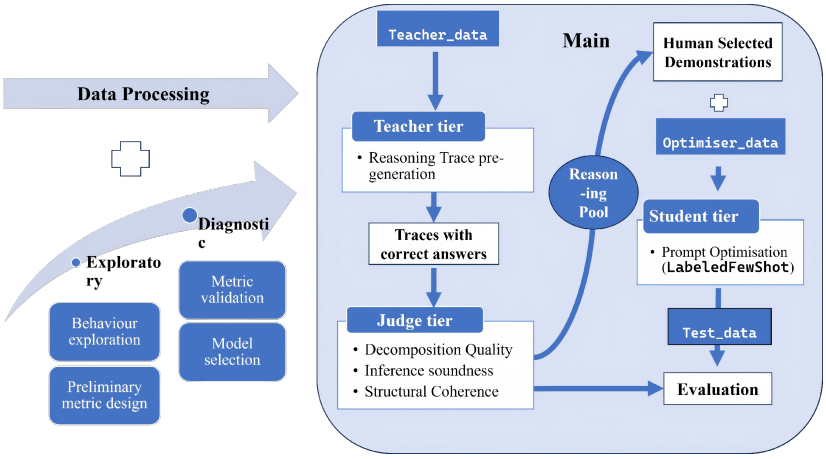


Figure 4.1: Research Design Overview

the *main* implementation for systematic instrument development and model selection. Outcomes of both the exploratory and diagnostic stages jointly constrained the models for deployment and the intermediate evaluation and selection mechanism, as visualised in Figure 4.1.

This three-stage design (i.e., *exploratory*, *diagnostic* and *main*) forms the structural foundation of this chapter. The following sections describe the role and objectives of each stage, discuss how exploratory observations and diagnostic refinements serve the pipeline, and establish the configuration for the main stage. The chapter concludes with methods for the final evaluation.

4.1 Exploratory Stage

At this stage, I conducted a lightweight experiment to uncover domain-specific model reasoning behaviour patterns, contributing to the definition of a targeted metric dimension.

4.1.1 Lightweight Experiment

To operationalise, I prompted the five available LLMs (i.e., GPT-4o-mini, GPT-4.1-mini, GPT-5.4-mini, O4-mini, and Deepseek-R1) with simply realistic task conditions below:

> *Solve the cryptic clue. Reason that it generally consists of two parts, the definition and the wordplay. The definition is directly related to the answer, like a synonym or hyponym, and the wordplay needs to be deciphered,*

usually with the help of an indicator that hints at what you need to do.

With this V0 prompt, all six models generated reasoning traces and final answers to the clues in the `preliminary_data`. In this process, gold annotations and answers were not passed to the models. Meanwhile, the temperature for non-thinking-LLMs (i.e., GPT-4o-mini, GPT-4.1-mini, and GPT-5.4-mini) was 0.6 to allow variance while maintaining consistency in the clue-solving process.

Then, to examine the behaviour of the models for designing the evaluation scheme, I scored each reasoning trace and the corresponding final answer generated based on final answer correctness and two generic fine-grained reasoning evaluation criteria suggested by Lee and Hockenmaier (2025): faithfulness and global correctness.

In parallel with the above scoring, I made qualitative observations of reasoning patterns per model, borderline cases, and potential directions for defining domain-specific evaluation criteria. The results of the experiments are presented in Appendix A. At this stage, observations were not treated as statistically generalisable findings but as formative insights to support the refinement of downstream instruments.

Moreover, to test the possibility of implementing LLM-as-a-Judge for the intended fine-grained evaluative scenario, I deployed GPT-4.1-mini with the criteria above as a judge to generate scores and justifications for the 40 reasoning traces generated by DeepSeek-R1. As shown in Appendix A, LLMs can be implemented to evaluate a reasoning trail to solve cryptic clues, although careful selection of the judge and calibration at the diagnostic stage are needed. How the LLM-as-a-judge mechanism was designed and implemented will be further described in Section 4.2.1.

4.1.2 Preliminary Metric Design

The results from the previous sub-section grounded the definition of three targeted dimensions in Table 4.1, forming the basis for subsequent instrument development.

Final accuracy was integrated as an additional dimension to measure whether the solver's final answer aligns with the gold answer. Although not part of the reasoning evaluation (Lightman et al., 2024), it provides an outcome-level reference: a reasoning process may appear coherent and well-structured while still leading to an incorrect solution. It also facilitates the filtering of teacher reasoning traces.

Dimension	Description & Justifications	Behaviour Examples
Decomposition Quality (DQ)	Whether the solver correctly identifies and segments components: *definition, wordplay, indicator*, and *letter count*. This is a cryptics-specific criterion. Both Sadallah et al. (2025) and the preliminary exploration suggest that complete surface-level decomposition is fundamental to the solving process.	• **Expected:** Correctly splits "Creature seen in our ocean frequently (4)" into *definition* ("Creature") and *wordplay*. • **Unexpected:** No clue segmentations or invented components that cannot be explicitly derived from the clue itself.
Inference Soundness (IS)	Whether the solver's reasoning trace is mechanically valid and internally reconstructible under standard cryptic clue conventions. This dimension assumes the premise (i.e., decomposed components) is fixed. Meanwhile, it conjugates 'validity' and 'faithfulness' from the universal criteria (Lee and Hockenmaier, 2025) given their observed interrelations.	• **Expected:** Explicit wordplay and letter manipulation inference with intermediate explanations. • **Unexpected:** Pure synonym substitution and significant logical leaps from the premise to the conclusion.
Structural Coherence (SC)	Whether the reasoning trace is stepwise consistent and globally well-structured, regardless of whether the final answer or intermediate inferences are correct. This dimension integrates 'coherence' and 'utility' suggested by Lee and Hockenmaier (2025), considering both structural incompleteness and thinking redundancy observed in model behaviour exploration.	• **Expected:** A clean chain of thought that utilises and entangles the intermediate clue decomposition and wordplay disambiguation to arrive at the final solution. • **Unexpected:** redundant reasoning, self-contradiction and information drift.

Table 4.1: Overview of the Evaluation dimensions.

4.2 Diagnostic Stage

As introduced earlier, the intermediate diagnostic stage aimed to constrain the models across tiers and how the evaluation dimensions proposed above should be operationalised. To achieve these goals, I first conducted a systematic metric calibration to establish reliable evaluation conditions and judge selection. Subsequently, I integrated the developed evaluation scheme to select teacher and student models. I will present the process and results of these steps in Sections 4.2.1 and 4.2.2, respectively.

4.2.1 Instrument Development

Since the paradigm presented in Figure 4.1 relies on curated traces for demonstration, well-defined boundary conditions are necessary to utilise the three fine-grained dimensions introduced in Section 4.1. To achieve an instrumental and reliable evaluation metric, I carried out iterative calibration with two judge-LLM candidates (i.e., GPT-4.1-mini and GPT-5.4-mini). GPT-4.1-mini was selected due to the wide deployment of the GPT-4 family in the LLM-as-a-Judge methodology (Huang et al., 2025). Meanwhile, GPT-5.4-mini was selected due to its recentness and its claimed performance in complex tasks (OpenAI, 2026).

Iterative Calibration

For calibration, I used the `preliminary_data` and the corresponding forty reasoning traces and answers generated by DeepSeek-R1 at the exploratory stage as a resource for pilot annotations. Across annotation rounds, a judge-LLM and I independently scored each trace with a version of the evaluation metric. The deployed LLM was also asked to generate justification for its evaluations.

I then compared the resulting annotations to analyse the judge-LLM's application consistency and alignment with my evaluations. Here, both a quantitative analysis of inter-annotator agreement and a qualitative manual inspection of disagreement were implemented. Moreover, since the primary goal of this metric development was trace selection, I inspected the overlap between the high-quality traces selected by the judge-LLM and me.

These insights jointly informed subsequent manual revisions of the evaluation metrics, including evaluation prompts, dimension descriptors, and scoring conditions. The revised versions were tested in a few rounds of annotation and comparison outlined above, which aimed to check evaluation reliability and selection functionality. In addition, I experimented with different judge-LLM candidates to investigate their evaluation capabilities and alignment with manual evaluations. Appendix B selectively presents the calibration process at the initial, middle, and final phases of metric development.

Consequently, this calibration process first defined the judge-LLM. To compare, the temperature for both models was set to 0.2. As compared in Appendix B.2.3, GPT-5.4-mini aligned better with my evaluation in terms of inference and structure of the reasoning traces generated, while GPT-4.1-mini aligned better in the dimension of clue decomposition. Given the importance of inference in solving cryptic clues (Efrat et al., 2021), I selected GPT-5.4-mini as the judge-LLM.

A few other calibration rounds thereafter yielded a stable evaluation metric in Table 4.2 on the next page.

When comparing the annotations using this metric, there was substantial agreement using Krippendorff's alpha as a benchmark (see Appendix B.3). Meanwhile, disagreement was not random but predominantly adjacent boundary cases, reflecting divergence in the annotators' decision thresholds (e.g., 1-2 rating differences). Moreover, a considerable overlap was observed in the top traces scored by GPT-5.4-mini and me. Therefore, these results indicate that the metric was functionally stable as a selection signal for high-quality reasoning traces within the judge tier. The following section substantiates the important design choices of this metric version.

Metric Design

The evaluation metric developed is rubric-based. Despite being interwoven in cryptic-clue solving, the three predefined dimensions (i.e., *Decomposition Quality* (DQ), *Inference Soundness* (IS), and *Structural Coherence* (SC)) are treated as independent domains. This separation considered the cross-dimensional performance divergence observed during exploration. For example, a model could perfectly segment the components while failing to infer a logic letter transformation. Therefore, this dimension-wise evaluation provides nuanced insight into the traces generated for trace selection, reflecting the notion of fine-grainedness (Lee and Hockenmaier, 2025).

Moreover, to quantify the evaluation process, I integrate a pointwise scoring approach by having the LLM-judge generate discrete scores for one reasoning trace per dimension (Li et al., 2025). To balance expressivity and consistency (Wang et al., 2025), the evaluation metric uses a binary (0–1) scale for SC due to its categorical nature. For the remaining two dimensions (i.e., DQ and IS), a tight three-point Likert scale (0–1–2) was adapted to distinguish partial decomposition or validity of inference.

Inspired by Lee et al. (2025), the metric design decomposes the evaluation dimensions into rule-based scoring instructions and descriptors, such as segmentation of *definition* and auxiliary components in DQ. In addition, scale descriptors operate under composite logical conditions (AND/OR) to determine whether a specific score should be awarded, reflecting the use of rule augmentation as a prompting strategy to improve evaluation accuracy. This practice is inspired by the common prompting strategies in the LLM-as-a-Judge methodology surveyed by Li et al. (2025).

Dimensions	What it measures	Scales
Decomposition Quality (0/1/2)	*Given the clue and the provided "useful information," evaluate and justify whether the given reasoning correctly identifies and segments the following components:**definition, wordplay, indicator (if present), and required number of letters**. Wordplay type classification belongs to inference, not decomposition.* • Only assess component identification and boundary detection, **not** the correctness of the underlying cryptic mechanism or final answer. • No exact case and string matching is required; partial or paraphrased identification is acceptable as long as the component role is correctly inferred. • If the reasoning identifies plausible definition and wordplay spans grounded in the clue text but explains the transformation incorrectly, (partial) credits should be awarded. • If multiple reasoning attempts are present (evidenced by 'However', 'Reassessing', 'Instead', and 'Alternatively'), evaluate the last stable segmentation that is explicitly used for the conclusion.	• **0**: Assigned when either: A) *Definition* is missing, incorrectly identified, or conflated. **OR** B) The decomposition is not supported by the clue text (i.e., hallucinated or arbitrary structure). Example: invented definition or indicator spans • **1**: A) The reasoning correctly identifies both definition and wordplay as distinct components, even if imperfectly segmented **AND** B) At least one of the auxiliary components is incorrect or missing: indicator misidentified/missing OR letter count incorrect / omitted OR weak/ambiguous mapping of auxiliary structure. Use 1 ONLY when the overall structure (definition + wordplay) is recoverable from the reasoning, but one or more components are missing, weakly specified, or not clearly grounded in the clue text. • **2**: Assign when definition, wordplay, and all relevant auxiliary components (indicator, fodder/material, required number of letters where available) are identified with sufficiently clear boundaries to reconstruct the intended structure, and no hallucinated components are introduced.
Inference Soundness (0/1/2)	*Evaluate and justify whether the reasoning trace is mechanically valid and internally reconstructable under standard cryptic clue conventions. The given 'useful_information' is used for reference only, since the routes to solving the wordplay can vary.* • Assume the decomposition is fixed. Do NOT evaluate the correctness of the definition and indicator identification! Focus on stepwise inference quality: GIVEN THE PREMISE, HOW IS THE INTERMEDIATE THINKING? • Do NOT repair, reinterpret, or replace the reasoning with alternative solutions. • If multiple reasoning attempts are present (evidenced by 'However', 'Reassessing' and 'Alternatively'), evaluate the last stable segmentation that is explicitly used for the conclusion.	• **0**: Assigned when either: A) The clue contains invalid cryptic logic. Examples: answer-first reasoning without a valid derivation, incorrect letter counts, pure guessing or synonym substitution, considering active wordplay components misdirected/useless, etc. **OR** B) cannot be mechanically followed **OR** C) The reasoning is not clue-based. • **1**: Assigned when: A) the reasoning is directionally plausible **AND** B) contains no more than two unclear, incorrect, or insufficiently supported steps. Examples: correct premise but incorrect inference, incomplete explanation of a transformation, partial but not fully justified letter manipulation • **2**: Assigned when the reasoning is fully mechanically valid and reconstructable. All transformations follow standard cryptic conventions and clearly support the final answer. Minor omissions are allowed only if the full derivation remains unambiguous and recoverable without guesswork.
Structural Coherence (0/1)	*Given the 'solver_reasoning', evaluate whether the reasoning trace is stepwise consistent and globally well-structured, regardless of whether the final answer or intermediate inferences are correct.* • The reasoning trace is fixed and must not be modified, repaired, or reinterpreted. • Do not penalise incorrect conclusions or invalid cryptic clue interpretations. A reasoning trace may still receive 1 if the reasoning process itself remains structurally coherent and internally consistent. • Focus exclusively on structural properties of the reasoning process, including progression, cross-step consistency, step utility/necessity, and whether the final answer is reached through a coherent construction process. • Do not penalise incorrect conclusions or invalid cryptic clue interpretations.	• **0**: Assign 0 when any of the following holds: A) Global structure gap: The reasoning lacks a clear and followable progression from clue analysis to final answer construction. This includes fragmented reasoning, missing or unclear decomposition of the clue, early arrival at an answer without sufficient intermediate support, or abrupt shifts in interpretation that break the flow of the solution path. **OR** B) Local inconsistency: The reasoning introduces contradictions, incompatible interpretations, abandoned analyses, or reversals that are not resolved. **OR** C) Weak construction: The intermediate reasoning steps are too vague, loosely specified, or not functionally connected to the final answer. **OR** D) Explicit failure signal: The solver explicitly admits uncertainty, inability to complete the solution, or failure to construct a valid answer path. • **1**: Assigned when the reasoning maintains a stable and followable solution path from decomposition to conclusion. Features include: Clear clue segmentation and decomposition, Coherent progression between reasoning steps, Consistency across intermediate steps, Steps that are necessary and meaningfully contribute to constructing the final answer, and a fully traceable construction path from clue to answer.

Table 4.2: Calibrated Evaluation Metric

4.2.2 Model Selection

In addition to the controlled evaluation and selection mechanism presented above, the main prompting paradigm required explicit pedagogical roles and model definitions for the teacher and student tiers.

In the current paradigm, an LLM is deemed a teacher when it can a) achieve higher final accuracy, and b) generate higher-quality reasoning traces for demonstration purposes compared to the other candidate models for the same clues in the `preliminary_data`. Conversely, a student LLM was characterised by a lower baseline performance than the teacher to allow room for improvement while still exhibiting limited but non-trivial cryptic reasoning, such as identifying *definitions* and *indicators*.

Although observations in the exploratory stage provided an understanding of divergence in model performance, a more controlled model selection process was necessary. To achieve this, I executed the judge-LLM (GPT-5.4-mini) with the metric developed in Section 4.2.1 to score the reasoning traces generated by the model candidates (i.e., Deepseek-R1, GPT-5.4, O4-mini, GPT-5.4-mini, GPT-4.1-mini, and GPT-4o-mini) under the same configuration in the exploratory stage. Final accuracy was also calculated. Table 4.3 provides an overview of model performance.

Models	**Acc**	μ_{DQ}	μ_{IS}	μ_{SC}
DeepSeek-R1	0.40	0.80	0.38	0.68
GPT-5.4	0.30	0.90	0.42	0.50
O4-mini	0.82	1.48	1.45	1.00
GPT-5.4-mini	0.17	0.60	0.20	0.40
GPT-4.1-mini	0.28	1.10	0.30	0.65
GPT-4o-mini	0.05	0.55	0.10	0.33

Table 4.3: Diagnostic Stage Model Selection

As a result, O4-mini outperformed the other candidates in final accuracy and reasoning quality on the `preliminary_data`, supporting its role as a teacher. Manual inspection of the traces further confirmed its clarity and explicitness in clue decomposition and inference. However, some traces exhibited a parallel decomposition of *definition* and *wordplay* without explicit interconnections (e.g., checking if the word derived from the letter transformation aligns with the definition segmented). This observation thus implied the need to refine the current teacher prompt regarding component interweaving, which will be presented in Section 4.3.1.

Additionally, GPT-4.1-mini was chosen as the student due to its lower but moderate reasoning and solving performance compared to the teacher's, aligning with the role defined above. GPT-5.4-mini could have been selected, but this could raise biases in

self-evaluation (i.e., using GPT-5.4-mini to evaluate the reasoning traces generated by the same model).

The results of the exploratory and diagnostic stages described above, together with the data processing in Chapter 3, determined the models, the judge mechanism, and the database. Together, they established the foundation for the main implementation of the prompting paradigm.

4.3 Main Implementation

As demonstrated in Figure 4.1, the adapted prompting paradigm encompasses three key tiers: teacher, judge, and student. These stages were executed independently to mitigate propagation. The subsections below detail the objective and implementation per layer.

4.3.1 Teacher Tier

To begin with, the teacher layer aimed to pre-generate reasoning traces on sampled clues across different wordplay types (i.e., `teacher_data`) for subsequent evaluation, selection and demonstration.

`Teacher_data` Given the imbalanced distribution in Figure 3.2 and the need to ensure the demonstration diversity across wordplay, 50 clues per wordplay type, including their annotations, were randomly sampled from the training set. As a result, this amounted to 400 pre-generation instances in `teacher_data`.

Model Configuration O4-mini was then called through DSPy (Khattab et al., 2023) to generate both intermediate reasoning steps and final answers to solve the clues in the `teacher_data`. Here, the model was exposed to neither the correct answer nor other clue-oriented annotations.

Meanwhile, I revised the V0 prompt (task description) used in the exploratory stage with domain expertise, as shown in Figure 4.2 below.

Compared to the basic version in the exploratory experiments, this task prompt featured more task-specific information (e.g., wordplay definitions) and generation constraints. Figure 4.3 demonstrated the reasoning traces before and after the prompt update to solve the clue "*Filmmaker to sensationalise hollow admission by scientist*

```
Solve the cryptic clue. Reason that it generally consists of two parts, the definition
and the wordplay.

The definition is directly extracted from the clue and related to the answer, like a
synonym or hyponym, and usually comes at the beginning or the end of the clue.
The wordplay needs to be deciphered, usually with the help of an indicator that hints at
what you need to do. Here is a list of possible wordplay types that may occur, and their
descriptions:
- Anagram: An anagram is a word (or words) that, when rearranged, forms a different word
or phrase.
- Insertion: An insertion is where you embed one word or letter sequence directly inside
another.
- Container: One word is placed inside another (or outside another) to get the answer.
- Reversal: taking a specific sequence of letters and spelling them backward.
- Hidden: the answer appears consecutively within part of the clue text.
- Deletion: letters are removed from a word or phrase.
- Homophone: the answer is indicated through a similar pronunciation.
- Alternation: letters are selected according to a pattern, such as every other letter.

Solve the cryptic clue with a complete, coherent, and concise step-by-step explanation.
Each step must apply a single clue component (indicator or fodder) to progressively
construct the answer.
The definition must not be treated as a separate analytical branch; it should be
extracted first without suggesting the answer AND also used as a final validation
constraint on the constructed result.

All reasoning must proceed as a single constructive chain driven by wordplay
transformations. All steps must remain strictly grounded in the clue text or standard
cryptic conventions; no step may introduce unsupported interpretations or
transformations.
```

Figure 4.2: Final Generation Prompt

(10)".

In producing the same final answer, the second reasoning trace is more explicit in letter transformation and more coherent in connecting the identified *definition* and the conclusion derived from the letter manipulations.

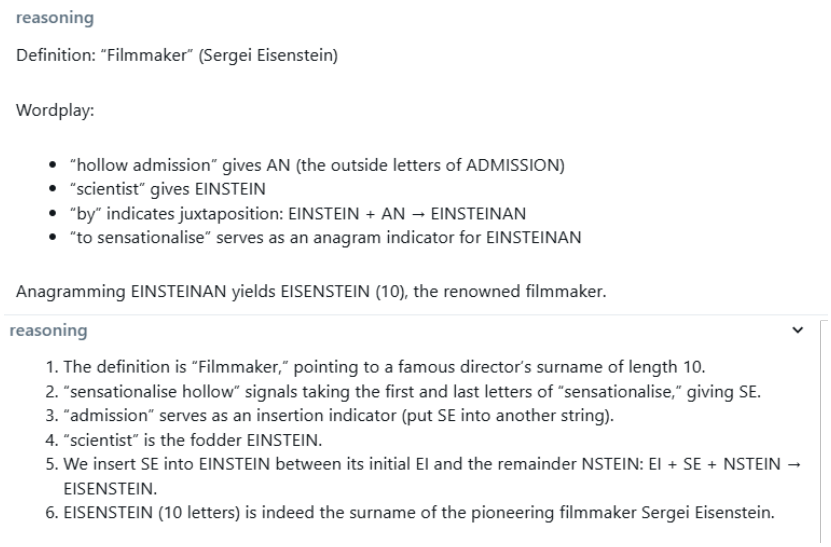


Figure 4.3: Reasoning Trace Comparison before (top) and after (bottom) the Prompt Revision on the clue: *'Filmmaker to sensationalise hollow admission by scientist* (10)'

Preliminary Answer-Based Filtering The teacher tier ended with a preliminary filtering that calculated the total final accuracy and retained reasoning traces with correct answers. This helped mitigate cases where the model could provide a seemingly convincing account despite incorrect answers. This allows a more conditioned evaluation and selection that focuses on trace quality.

4.3.2 Judge Tier

The intermediate judge tier uses the fine-grained evaluation scheme and GPT-5.4-mini at the diagnostic stage to evaluate and select potential quality reasoning traces for demonstrations. In this process, in adherence to the notion of dimension separation in metric design justified in Section 4.2.1, each call of the judge-LLM was aimed at one dimension only.

After the three rounds of evaluation above, the scores for each reasoning trace were automatically summed. For each wordplay type, the top-10 reasoning traces were automatically selected to form a reasoning pool as the primary source of the student prompt.

4.3.3 Student Tier

In the student layer, the student LLM (GPT-4.1-mini) was prompted with demonstrations in the reasoning pool for optimisation. However, the prompting challenges discussed in Section 2.3.2 imply the need for prompt optimisation concerning demonstration selection and ordering. I thus conducted a semi-automatic optimisation.

Semi-automatic Prompt Optimisation Following the practice implied by McDonald and Emami (2024), I played a proactive role in the curation process. I manually picked one reasoning trace per wordplay category that I assumed was demonstrative among the ten candidates from the automatic selection procedure above.

Subsequently, I tested the student-LLM with different ordering of hand-picked demonstrations on the `optimiser_data`. To automate this process, I used `LabeledFewShot` optimiser from DSPy (Khattab et al., 2023). Here, the ordering that allowed the student-LLM to achieve the best performance was assumed to be the optimised demonstration set. The best ordering from this experiment was integrated into the task description in Figure 4.2.

Chapter 5 will present in detail the results of the key steps in the main implementation, including the final set of demonstrations.

4.4 Evaluation

The performance of the curated demonstration is evaluated in twofold. It entails a quantitative performance comparison and a qualitative reasoning analysis. This dual evaluation aims to investigate the effectiveness of adding curated reasoning demonstrations not only against standard CoT prompting (Sub-RQ1) but also in terms of distilled reasoning structures and strategies (Sub-RQ2). For interpretation references, the teacher model was prompted under the pre-generation configuration to generate intermediate reasoning steps and the final answer.

4.4.1 Quantitative Performance Evaluation

To establish the basis for evaluation, a baseline (prompted with the standard zero-shot CoT) and the optimised student-LLM independently generated intermediate reasoning steps and final answers for the held-out `test_data` with the same task description.

Each reasoning trace was subsequently scored by the judge-LLM with the developed metric, whose results served as the primary resource for quantitative comparison. To compare both at the general and wordplay levels, the corresponding final accuracy and the averaged reasoning quality scores were automatically calculated. This helped answer Sub-RQ1.

4.4.2 Qualitative Reasoning Analysis

I manually compared and analysed the reasoning traces that were scored differently. Here, I conducted a tripartite comparison among the baseline, the optimised, and the teacher traces for the targeted clues. This practice aimed to identify any emergent reasoning patterns that contributed to performance improvement and/or degradation with respect to Sub-RQ1. This analysis also helps answer what is learned after the demonstration (Sub-RQ2).

In addition, for the dimensions of decomposition quality and inference soundness, I particularly focused on cases where the target clue shared the same component or mechanical distribution as those in the demonstration set for analysis. This helps examine the effectiveness of the demonstrations from another perspective.

Chapter 5

Implementation

Following the research design outlined in Chapter 4, I first completed the procedures for the main implementation of the prompting paradigm, including the formation of an a priori reasoning pool, demonstration hand-picking and ordering optimisation. This chapter briefly reports on the results for each step.

5.1 Reasoning Pool Formation

Within teacher tier, O4-mini was prompted to generate reasoning traces and final answers to the 400 clues in the `teacher_data`. Since the clue '*Regularly fail sick friend in financial matters* (8)' violates the Azure content filter, I removed it from the `teacher_data` for accuracy calculation. As a result, the teacher-LLM achieves a final accuracy of 0.64, resulting in 255 of 399 corresponding reasoning traces for subsequent evaluation.

As shown in Table 5.1, O4-mini produced correct answers mainly for clues encoding the wordplay of 'hidden' (41) and 'anagram' (39), while it answered only half of the clues encoding homophones in the `teacher_data` correctly.

These 255 traces were then passed to the fine-grained LLM-as-a-Judge mechanism within the judge layer. Table 5.1 outlines the conditioned means per wordplay type, as well as for all clues. In general, the teacher-LLM outputted moderate-to-high-quality reasoning traces with correct answers. Notably, it achieved a near-perfect structural coherence ($\mu_{SC|correct}$= 0.996). Meanwhile, the teacher-LLM achieved a substantial level of inference soundness, with the corresponding overall means exceeding 1.5 out of 2.

Subsequently, with the top-10 thresholding methods for automated trace selection, a reasoning pool with 80 trace candidates for demonstrations was formed. The full pool

Wordplay	**Nr. of Traces**	$\sum \mu$|**corr**	μ_{DQ}|**corr**	μ_{IS}|**corr**	μ_{SC}|**corr**
Anagram	39	4.538	1.641	1.897	1.000
Insertion	32	3.969	1.469	1.500	1.000
Container	30	4.333	1.567	1.767	1.000
Reversal	29	3.862	1.448	1.448	0.965
Hidden	41	4.171	1.610	1.560	1.000
Deletion	27	3.815	1.370	1.444	1.000
Homophone	25	4.520	1.720	1.800	1.000
Alternation	32	4.656	1.750	1.907	1.000
All	**255**	**4.243**	**1.576**	**1.670**	**0.996**

Note: All averages are conditioned exclusively on traces that achieved correct answers.

Table 5.1: Final accuracy and average reasoning quality scores conditioned on correct answers for teacher pre-generation

can be found in the repository. This served as the basis for the subsequent selection and optimisation for student-prompting.

5.2 Student-LLM Prompting

As described in Section 4.3.3, I selected eight reasoning demonstrations from the above-generated reasoning pool and tested them with different orderings through different random seeds. Table 5.2 presents the performance of GPT-4.1-mini with different demonstration orderings on the `optimiser_data`.

Seeds	**Acc**	μ_{DQ}	μ_{IS}	μ_{SC}
42	0.290	0.965	0.280	0.410
43	0.345	1.010	0.240	0.365
44	0.330	0.955	0.265	0.425
45	0.340	1.015	0.310	0.400
46	0.285	0.970	0.260	0.425

Table 5.2: Ordering Optimisation Experiments

The performance in Table 5.2 signals that ordering can cause performance variance in GPT-4.1-mini. Given that performance was relatively optimal across all dimensions with the seed of 45, the curated demonstration set presented on the next two pages was ultimately augmented to guide the target student-LLM to solving cryptic clues.

```
{
        "clue": "Calypso house has lively area (4)",
        "reasoning": "1. The definition is at the end of the clue:
        ↪  \lively area."  \n2. The wordplay indicates a hidden answer
        ↪  (\has" signals that the answer is contained in the
        ↪  preceding words).  \n3. Look inside \CALYPSO HOUSE" for a
        ↪  consecutive string of four letters: ...CALYP[SOHO]USE...
        ↪  \n4. Those four letters spell SOHO.  \n5. SOHO is famously
        ↪  a lively district (area), matching the definition.",
        "answer": "SOHO"
      },
      {
        "clue": "Sin that's petty with leader going, penning personal
        ↪  resignation (8)",
        "reasoning": "We need an 8-letter word meaning \Sin" (the
        ↪  definition at the beginning). The rest is
        ↪  wordplay:\r\n\r\n1. petty → TINY  \r\n2. \with leader
        ↪  going" → remove the leader (first letter T) from TINY,
        ↪  giving INY  \r\n3. personal resignation → I QUIT (I =
        ↪  personal pronoun; QUIT = resign)  \r\n4. \penning" →
        ↪  container indicator, so insert IQUIT into INY  \r\n   IN +
        ↪  IQUIT + Y → INIQUITY  \r\n5. INIQUITY means \sin," matching
        ↪  the definition.",
        "answer": "INIQUITY"
      },
      {
        "clue": "Ceremony is proper for the audience (4)",
        "reasoning": "1. Definition: \Ceremony" suggests the answer is
        ↪  a word meaning ceremony.  \r\n2. Wordplay indicator: \for
        ↪  the audience" signals a homophone.  \r\n3. Fodder: \proper"
        ↪  gives RIGHT.  \r\n4. Homophone of RIGHT → RITE.  \r\n5.
        ↪  RITE (4 letters) matches the definition \ceremony."",
        "answer": "RITE"
      },
      {
        "clue": "Mexican food thus, queen perhaps served up? (5)",
        "reasoning": "1. Definition: \Mexican food" (5) suggests TACOS.
        ↪  \r\n2. Wordplay: \thus" → SO.  \r\n3. \queen perhaps" → CAT
        ↪  (a \queen" is a breeding female cat in animal-breeding
        ↪  jargon).  \r\n4. Together \thus, queen perhaps" gives SO
        ↪  CAT.  \r\n5. \served up?" is a reversal indicator. Reverse
        ↪  SO CAT → TACOS.  \r\n6. TACOS matches the definition
        ↪  \Mexican food."",
        "answer": "TACOS"
      },
      {
        "clue": "Doctor stationed to provide remedies (9)",
```

```
      "reasoning": "1. Definition: \provide remedies" indicates a verb meaning \gives remedies."  \r\n2. \Doctor" is the anagram indicator.  \r\n3. Fodder: the nine letters of \stationed."  \r\n4. Anagramming STATIONED → ANTIDOTES.  \r\n5. ANTIDOTES means \provides remedies," matching the definition.",
      "answer": "ANTIDOTES"
    },
    {
      "clue": "Accept offer to go into A&E (5)",
      "reasoning": "1. The definition is \Accept," which can be clued by the verb ABIDE (as in \abide by the rules" = \accept the rules").  \r\n2. The wordplay is \offer to go into A&E."  \r\n3. \Offer" cryptically clues BID.  \r\n4. \A&E" supplies AE (dropping the ampersand).  \r\n5. \To go into" signals insertion: insert BID into AE, yielding A + BID + E = ABIDE.  \r\n6. ABIDE means \accept," matching the definition and the 5-letter enumeration.",
      "answer": "ABIDE"
    },
    {
      "clue": "Rocky somewhere in Bedfordshire, initially lost (8)",
      "reasoning": "1. The definition is \Rocky," suggesting a word meaning unsteady or unstable.  \r\n2. \Somewhere in Bedfordshire" gives the place name DUNSTABLE.  \r\n3. The indicator \initially lost" tells us to remove the first letter of DUNSTABLE.  \r\n4. Dropping the initial D from DUNSTABLE yields UNSTABLE.  \r\n5. UNSTABLE (8 letters) means \rocky," confirming the definition.",
      "answer": "UNSTABLE"
    },
    {
      "clue": "Confounded figure out on bail regularly (8)",
      "reasoning": "1. The definition is \Confounded," which can colloquially mean \infernal" (as in \that infernal contraption").  \r\n2. Wordplay:\r\n   a. \figure" clued by INFER (to figure out).  \r\n   b. \on bail regularly" means take the letters in the even positions of the phrase ON BAIL → positions 2, 4, 6 give N A L.  \r\n3. Append NAL to INFER to build the answer: INFER + NAL → INFERNAL, matching the definition.",
      "answer": "INFERNAL"
    }
  ]
}
```

Chapter 6

Evaluation Results & Analysis

After implementing the demonstration configurations described in Chapter 5, the resulting optimised student-LLM (OS) was compared to the baseline configuration (BS) on the `test_data` using the evaluation methods designed in Chapter 4. This chapter aims to present and analyse the evaluation results in light of the two sub-RQs: performance differences across wordplay categories (Sub-RQ1) and the emergent reasoning patterns after demonstrations (Sub-RQ2).

6.1 General Performance

To compare the difference in student performance with curated demonstrations, the reasoning traces generated by BS and OS were evaluated using the evaluation scheme within the judge tier (i.e., final accuracy (Acc), decomposition quality (DQ), inference soundness (IS), and structural coherence (SC)). Table 6.1 presents the general accuracy and means of the dimension of reasoning evaluation per model version.

In general, OS achieved lower performance than BS in terms of final accuracy and across all dimensions of reasoning quality. However, there is a divergence in the degree of degradation. The change in the overall averaged DQ is almost negligible. This is in striking contrast to SC due to an approximate decrease of 26.4%. In comparison, degradation in IS is more moderate, from 0.367 to 0.308.

To further compare performance at the micro-level, average scores for each word-

Model	**Acc**	μ_{DQ}	μ_{IS}	μ_{SC}
BS	0.300	1.038	0.367	0.625
OS	0.286 (↓)	1.006 (↓)	0.308 (↓)	0.460 (↓)

Table 6.1: General Model Performance

play category were calculated. Meanwhile, I also extracted reasoning pairs with score differences. Hence, in the following sections, I will quantitatively compare performance changes across wordplay types per evaluation dimension. For each of the dimensions of reasoning quality (i.e., DQ, IS, and SC), I also qualitatively compare and analyse the reasoning pairs extracted.

6.2 Final Accuracy

As introduced earlier, the final accuracy (FA) serves as an additional preliminary outcome check. It compares whether the final answer generated by the models aligns with the gold answer. As a result, Table 6.2 presents the FA that BS and OS achieved per wordplay category.

Wordplay Category	**Nr.**	**BS**		**OS**	
		Acc	**Correct**	**Acc**	**Correct**
Anagram	62	**0.516**	**32**	0.484	30
Insertion	62	0.210	13	**0.226**	**14**
Container	62	**0.274**	**17**	0.258	16
Reversal	62	**0.177**	**11**	**0.177**	**11**
Hidden	62	**0.274**	**17**	0.258	16
Deletion	62	0.226	14	**0.242**	**15**
Homophone	62	**0.226**	**14**	**0.226**	**14**
Alternation	62	**0.500**	**31**	0.419	26

Table 6.2: Final Accuracy and Number of Correct Cases per Wordplay

Among all the wordplay categories, both models achieve the highest accuracy for ‘anagram’ and ‘alternation’, despite a small decrease in accuracy (0.032 and 0.081) after the demonstrative configuration.

The changes for the remaining classes vary. In particular, there is an increase in accuracy of 0.016 for both ‘insertion’ and ‘deletion’. In contrast, there is a slight decrease in accuracy for ‘container’ and ‘hidden’, both from 0.274 to 0.258. FA for the other classes, namely ‘reversal’ and ‘homophone’, remains unchanged. Thus, these results suggest a mixed effect of curated demonstrations on FA across wordplay, though the effect is not significant.

6.3 Decomposition Quality

As defined earlier in Section 4.1, Decomposition Quality (DQ) in the evaluation metric focuses on whether a model can correctly identify and segment the *definition*, *word-*

play, and other auxiliary components (e.g., *indicator* and *enumeration*) in a cryptic crossword clue. This dimension is independent of the inference correctness.

6.3.1 Quantitative Comparison

Following the notion above, reasoning traces generated by BS and OS were scored and compared. The average scores and the number of cases with a score change per wordplay were automatically calculated, as shown in Table 6.3.

Wordplay	**Nr.**	**BS-μ_{DQ}**	**OS-μ_{DQ}**	**+**	**-**	**=**
Anagram	62	**1.193**	**1.193**	**9**	**9**	44
Insertion	62	**1.016**	0.935	8	**12**	32
Container	62	**1.081**	0.887	4	**16**	42
Reversal	62	**0.887**	0.870	7	**8**	47
Hidden	62	0.967	**1.016**	**15**	11	36
Deletion	62	**0.935**	0.854	9	**15**	38
Homophone	62	1.000	**1.080**	**13**	9	40
Alternation	62	**1.226**	1.210	7	**8**	47

Table 6.3: Average Decomposition Quality Scores and Number of Score Changes per Wordplay

As a result, the average DQ remains the highest for ‘alternation’ and ‘anagram’. After the demonstration, there is a slight decrease in the student clue segmentation for ‘Alternation’ on average, while that of ‘anagram’ clues remains unchanged.

Moreover, there are varied changes in the other wordplay categories. When it comes to clues encoding ‘homophone’ and ‘hidden’, the corresponding average DQ increases by 0.080 and 0.049, respectively. This is likely due to their greater number of increased instances, which could suggest the positive impacts of demonstrations on DQ for these classes. However, the other classes witnessed a decrease in DQ, particularly for those related to ‘container’ clues (16 instances), indicating a mixed demonstrative effect on clue decomposition.

6.3.2 Qualitative Comparison

To first understand whether the improvement in ‘homophone’ represents a consistently positive decomposition strategy transfer from the demonstrations, I focused on instances where the clues share the same componential structure as the demonstrative example. Specifically, the demonstrative clue, ‘*Ceremony is proper for the audience* (4)’, follows a *definition* + *wordplay* + *indicator* pattern. Consequently, five corre-

sponding BS-OS reasoning pairs showing score differences were manually extracted and inspected.

Contrary to my expectations, the extracted pairs reflect both positive and negative score differences in DQ, indicating an uneven performance after the demonstration. These are demonstrated in (2) and (3), respectively:

(2) Clue: '*Dancer from dance scene for the audience* (9)'

- BS (DQ = 1):

> The clue is "Dancer from dance scene for the audience (9)".
> Step 1: Identify the definition.
> The definition is likely "Dancer" since it's at the start of the clue and fits the enumeration (9 letters).
> Step 2: Analyze the wordplay.
> The phrase "from dance scene for the audience" suggests a homophone (indicated by "for the audience").
> Step 3: Find a phrase or word related to "dance scene" that, when heard aloud, corresponds to a 9-letter word meaning "Dancer".
> ...
> Step 29: The phrase "dance scene" for the audience means to take "dance scene" and find a homophone for "ballerina".
> Step 30: "Ball arena" sounds like "ballerina".
> Step 31: "Dance scene for the audience" = "ball arena" (a place for dance) sounds like "ballerina" (a dancer).
> Step 32: So, the answer is "BALLERINA" (9 letters), meaning "Dancer".
> Step 33: Therefore, the wordplay is a homophone ("for the audience") of "ball arena" ("dance scene") = "ballerina" (Dancer).

- OS (DQ= 2):

> 1. The definition is likely "Dancer," referring to a type of dancer.
> 2. "from dance scene for the audience" suggests a homophone clue indicated by "for the audience."
> 3. "dance scene" could be the word "ballroom" or simply "ball" (a dance event).
> 4. A 9-letter word for a dancer is "BALLERINA."
> 5. The phrase "dance scene" sounds like "ball arena" when spoken aloud, which is a homophone for BALLERINA.
> 6. Therefore, the answer is BALLERINA, a dancer, derived by homophone wordplay from "ball arena" (dance scene).

(3) Clue: '*Former tennis champ put on the scales, we hear* (4)'

- BS (DQ = 1)

> The clue is "Former tennis champ put on the scales, we hear (4)".
> Step 1: Identify the definition.
> The phrase "Former tennis champ" is likely the definition, as it is at the start of the clue and fits the enumeration (4 letters).
> Step 2: Analyze the wordplay. The phrase "put on the scales, we hear" suggests a homophone ("we hear") of a word related to "put on the scales."
> ...
> Step 27: Could the answer be "Lley" (Lleyton Hewitt)?
> "Lley" sounds like "lay" (put down), which is related to "put on the scales" (put down weight).
> "Lley" is 4 letters.
> Lleyton Hewitt is a former tennis champ.
> So "Lley" fits the definition.

Step 28: So the answer is ”Lley,” derived from the homophone of ”lay” (put on the scales = lay down weight) and ”we hear” indicates homophone.
Therefore, the answer is ”Lley.”

- OS (DQ = 0)

1. The definition is likely “put on the scales,” suggesting a word meaning to weigh.
2. “We hear” indicates a homophone.
3. “Former tennis champ” could be a famous player’s name or nickname.
...
Another approach: “Former tennis champ” = “Edger” (no), “Roddick” (no), “Sam” (Sampras), “Pete” (Sampras). 22. “Pete” sounds like “peat” (no).
23. “Put on the scales” = “Weigh.” Homophone indicator “we hear.”
24. The answer is “Whey,” a homophone of “weigh,” 4 letters.
25. “Former tennis champ” is the definition or wordplay? Maybe “Former tennis champ” = “W” (initial letter) + “hey” (an interjection).
26. Possibly “Whey” is the answer, as it sounds like “weigh,” meaning “put on the scales.”
27. So the answer is WHEY.

Both examples showed that OS becomes more explicit and concise in the early identification of key clue components. Compared with the teacher’s traces below in (4) on the same clue as (2), ‘*Dancer from dance scene for the audience* (9)’, the degree of conciseness is notably similar. Therefore, I speculate that OS has learnt a stylistic pattern for explaining the decomposition process more clearly at the surface level.

(4) Clue: ‘*Dancer from dance scene for the audience* (9)’

1. Definition: “Dancer” at the start of the clue signals the definition of the solution.
2. Homophone indicator: “for the audience” suggests we need a word that sounds like another phrase.
3. Wordplay fodder: “dance” = BALL and “scene” = ARENA, giving the phrase BALL ARENA.
4. Apply the homophone: BALL ARENA (spoken aloud) yields BALLERINA.
5. Confirm length: BALLERINA is 9 letters, matching the enumeration.

However, a stylish alignment does not guarantee DQ across all clues. For instance, it identifies the definition at the start (‘dancer’) in (2) but in the middle (i.e., ‘put on the scales’) for (3), suggesting a limited transfer of stable boundary detection from the demonstration. Moreover, OS remains weak in the auxiliary components (e.g., *indicator* and *fodder*) after the demonstration. Similarly to the behaviour of BS, it still shifts away from clue-based decomposition and introduces unsupported speculations towards the final attempt, such as the one demonstrated in (3). This means the student’s decompositional behaviour can be instance-contingent.

This speculation is further evident when inspecting 20 reasoning pairs with score differences for the ‘container’ class. Consider (5a) and (5b) below, for instance:

(5a) Clue: '*Local is unaffected about temperature* (6)'

- BS (DQ = 1)

> The clue is "Local is unaffected about temperature (6)".
> Step 1: Identify the definition.
> The definition is likely "Local," which is a straightforward synonym clue.
> Step 2: Analyze the wordplay.
> The rest of the clue is "is unaffected about temperature."
> Step 3: Break down the wordplay.
> "unaffected" can be a word or a base word from which we derive the answer. "about temperature" suggests placing a letter representing "temperature" inside or around a word meaning "unaffected."
> ...
> Step 5: Try "UNFAZED" which means unaffected. "UNFAZED" is 7 letters, so no.
> Step 6: Another possible synonym for "unaffected" is "STABLE," which is 6 letters.
> Step 7: "STABLE" can mean "unaffected," and "Local" is the definitios
> ...
> Step 9: Try "NATIVE" which means local (6 letters).
> Step 10: Does "NATIVE" fit with the wordplay "is unaffected about temperature"?
> - "Unaffected" could be "NATIVE" because natives are local and unaffected by outside influence.
> - "About temperature" could mean inserting "T" (temperature) into a word meaning unaffected.
> ...
> par Step 19: Reconsider "Local" as the definition and "unaffected about temperature" as wordplay.
> Step 20: "Is unaffected" could be "NATIVE" (local, original, unaffected by change).
> Step 21: So, the answer is "NATIVE."

- OS (DQ = 2)

> 1. Definition is likely "Local."
> 2. "unaffected" suggests the word UNAFFECTED or a synonym, but we need 6 letters.
> 3. "about temperature" indicates a container or insertion with T (common abbreviation for temperature).
> 4. "unaffected" can be "INNATE" or "UNTINT" but these don't fit well.
> 5. Another approach: "unaffected" = NATURAL (7 letters, too long).
> 6. Consider "unaffected" = "INERT" (5 letters, too short).
> 7. Try "about temperature" = putting T inside a word meaning unaffected.
> 8. "Local" as definition means a word like "NATIVE," which is 6 letters.
> 9. "Unaffected" = NATIVE (meaning indigenous, local, and unaffected in a sense).
> 10. "about temperature" = T inside "naive" (naïve means unaffected).
> 11. Insert T into NAIVE: NA + T + IVE = NATIVE.
> 12. NATIVE means "Local," matching the definition.

(5b) Clue: '*Excited cheering as point is secured by Arsenal's neighbour* (9)'

- BS (DQ = 1)

> First, identify the definition part of the clue. The clue is "Excited cheering as point is secured by Arsenal's neighbour (9)". The definition is likely at one end, and "Excited cheering" fits well as a definition for a 9-letter word.
> Next, analyze the wordplay: "as point is secured by Arsenal's neighbour".
> - "Arsenal's neighbour" likely refers to a football club located near Arsenal FC. Arsenal is based in North London, and a well-known neighbour club is "Tottenham".
> - The word "point" in cryptics often refers to a compass point, such as N, E, S, or W.

- The phrase ”point is secured by Arsenal’s neighbour” suggests inserting a point (compass direction) inside the name of Arsenal’s neighbour. Tottenham has 9 letters: T O T T E N H A M We need to insert a point inside ”Tottenham” to get a word meaning ”Excited cheering”.
Try inserting ”E” (East) into Tottenham:
T O T T E N H A M
Insert E somewhere:
T O T T E N H A M already has an E, but maybe the insertion is more figurative.
. . .
Now, the clue says ”Excited cheering as point is secured by Arsenal’s neighbour (9)”. ”Excited cheering” is the definition. ”as point is secured by Arsenal’s neighbour” is the wordplay.
So, ”as” + ”point” inside ”Arsenal’s neighbour”. If the point is ”N”, then ”as” with ”N” inside becomes ”ans”.
Now, ”Arsenal’s neighbour” is ”Tottenham”. If we insert ”ans” inside ”Tottenham” or combine ”ans” and ”Tottenham” in some way.
. . .
Alternatively, ”enthusing” means excited cheering. So the answer is ”enthusing”.
”as point is secured by Arsenal’s neighbour” could be parsed as: point = N (north), secured by Arsenal’s neighbour (Tottenham).
So, ”en” (point N) inside ”thus” or ”thus” + ”ing” = enthusing.
Alternatively, ”enthusing” is the answer meaning ”excited cheering”.
Therefore, the answer is ENTHUSING.

- OS (DQ = 0)

1. Definition: Likely “Excited” at the start of the clue.
2. Wordplay: “cheering as point is secured by Arsenal’s neighbour”
3. “Arsenal’s neighbour” refers to a football rival located near Arsenal in London — likely “Spurs” (Tottenham Hotspur).
4. “point” in cryptic clues often indicates a compass point, such as N, E, S, or W.
5. “cheering as” is the fodder for an anagram indicated by “excited” (an anagram indicator).
6. Insert “point” (likely N) into an anagram of “cheering as” or part of it, or more likely, the entire phrase is an anagram of “cheering as” + N.
7. Letters in “cheering as” plus N: C H E E R I N G A S + N = 11 letters, but answer length is 9 letters.
. . .
31. Alternatively, “Arsenal’s neighbour” is “Spurs,” and the answer is “EAGERNESS” or “EXCITING.”
32. “Exciting” (9 letters) is a synonym of “excited.”
33. “Cheering as” anagrammed plus “point” (I or N) inside may give “EXCITING.”
34. Letters in “cheering as” = C H E E R I N G A S, rearranged plus or minus a point letter.
35. “Exciting” letters: E X C I T I N G (8 letters). Need 9 letters.
. . .
52. Anagram “CHEERING AS” = “EAGER CHINS,” “SEARCHING E,” “REACHING SE,” “ENRICH GAS E” no.
53. The only sensible word is “SEARCHING” (9 letters), which is an anagram of “CHEERING AS.”
54. “Searching” can mean excited or eager to find something, loosely “excited.”
55. Therefore, the answer is SEARCHING.

The wordplay chunk ‘unaffected about temperature’ in (5a) reflects a binary container construction of ‘*indicator* + elements to insert/contain’, similar to the clue in the demonstration ‘Calypso house has lively area (4)’. As a result, by imitating the decomposition procedure and the explanation style in the demonstration, OS becomes more stable in its segmentation than BS, which accounts for the increased score. This

change can be consistently observed in all four extracted increased cases.

However, the wordplay chunk 'Excited cheering as point is secured by' in (5b) involves a multi-component construction, requiring prior isolation of multiple fodder elements. This increases the ambiguity of the segmentation. Moreover, identifying the *definition* of 'Arsenal's neighbour' in (5b) poses an extra challenge, since it requires knowledge related to British culture. As a result, rigid imitation of the same procedures for those with explicit segmentation signals, such as the one in (5a), without understanding of the actual structure, is insufficient. The decomposition remains tentative compared to BS in (5b). This notion is evident in the 16 pairs of reasoning traces with a negative difference in scores.

Consequently, the analysis above reveals that the student-LLM might have learned the shallow decomposition procedures without really understanding what impacts the strategy choice of multi-fodder segmentation in the demonstrations. Compared to the standard zero-shot COT on the sME instance, the effectiveness of the curated demonstration on DQ is not only wordplay-dependent but also clue-contingent.

6.4 Inference Soundness

Inference soundness (IS) in the developed evaluation metric targets the mechanical validity and reconstructability of the intermediate reasoning steps, given the components decomposed as premises.

6.4.1 Quantitative Comparison

Table 6.4 presents the average scores of IS per wordplay category. As a result, the highest and lowest averages are still for 'anagram' and 'reversal', respectively. This suggests that logical inference for clues encoding 'reversal' remains generally challenging to the student-LLM after the demonstration.

Regarding changes, the IS average scores decrease for most wordplay categories presented. In particular, there is a decrease of 0.113 in the averages for 'anagram', 'deletion', and 'alternation', suggesting more salient negative post-demonstration inference patterns for these categories. Exceptionally, there is a slight increase in the average IS score for 'Container' from 0.274 to 0.316, while the score for 'insertion' remains unchanged.

Wordplay	Nr.	BS-μ_{IS}	OS-μ_{IS}	+	-	=
Anagram	62	**0.677**	0.564	3	**9**	50
Insertion	62	**0.210**	**0.210**	4	**5**	53
Container	62	0.274	**0.306**	**7**	**7**	48
Reversal	62	**0.129**	0.096	1	**3**	58
Hidden	62	**0.435**	0.370	6	**7**	49
Deletion	62	**0.258**	0.145	5	**10**	47
Homophone	62	**0.355**	0.290	5	**8**	49
Alternation	62	**0.597**	0.484	6	**10**	46

Table 6.4: Average IS Scores per Wordplay and Number of Score Changes per Wordplay

6.4.2 Qualitative Comparison

To understand the effects of the curated demonstrations on the IS of the student LLM, 57 reasoning pairs showing score changes related to 'anagram', 'alternation', 'deletion', and 'container were automatically extracted for a targeted analysis.

Across 12 out of 20 reasoning pairs with a positive score difference, I observed that OS becomes more deterministic and consistent in applying letter manipulation based on cryptic conventions when the target clue encompasses one wordplay signal, as demonstrated in (6):

(6) Clue: '*A villain dropping old dispute* (5)'

- BS (IS = 0)

> ...The wordplay involves 'A villain dropping old'. 'A villain' could be a word like 'EVILIAN' or 'VILLAIN', but since the clue suggests dropping 'old', we consider the abbreviation for 'old' which is 'O'. Removing 'O' from a word related to villain might give the answer. 'A villain' could be 'VILLAIN'. Dropping 'O' from 'VILLAIN' does not work directly, so consider a synonym for villain: 'FIEND'. Dropping 'O' doesn't apply here. ...Perhaps the definition is 'A villain dropping old' meaning the answer is a word meaning 'dispute'. The answer is 'FEUDS' (plural of feud), which means disputes. So the answer is FEUDS.

- OS (IS = 1)

> ...
> 3. "Dropping old" suggests removing the letter O (common abbreviation for old) from a word meaning villain.
> 4. A 6-letter word for villain is "RIVAL" + O = "RIVALO" (not a word), so try another approach. 5. Another word for villain is "ROGUE" (5 letters), dropping O (old) leaves "RUGE" (not a word).
> 6. Consider "RIVAL" as "dispute," which fits the definition and length.
> 7. "A villain dropping old" means remove O from "RIVAL" plus O? No.
> ...
> 14. The definition is "dispute," so the answer is ARGUE.
> 15. Possibly, "A villain dropping old" is wordplay for ARGUE: "A (a) + ROGUE (villain) dropping O (old) gives R GUE, which with A gives ARGUE."
> 16. So, ROGUE (villain) dropping O = RGUE, adding A gives ARGUE.

17. ARGUE means “dispute,” matching the definition and length.

In the above example, the clue contains a clear wordplay signal indicated by ‘Dropping old’. Despite identifying this mechanism at the early stage, BS completely abandons wordplay analysis in the final inference step. Instead, it semantically guesses an answer ‘FIEND’ that fits the definition ‘Villain’. Consequently, no explicit cryptic mechanism supports the final answer, resulting in a score of zero for inference soundness.

In contrast, OS continues to frame the solution process around the identified deletion operation in the final inference. Although the derivation remains largely speculative and unsupported (e.g., introducing ‘disput’ as an answer and examining the related wordplay), the final answer is nevertheless linked to an explicit mention of the letter removal, accounting for its partial credit.

This behavioural change is not consistently observed in clues that encode multiple wordplay mechanisms, such as the one in (7):

(7) Clue: ‘*Crooked group I missed facing French street* (6)’

- OS (IS = 0)

...
2. “Group I missed” suggests removing the letter I from a word meaning group. A common word for group is “GANG.” Removing I is not possible here, so try another word: “GANG” does not contain I, so consider “GANGI” or “GANGI” is not a word. Another common word for group is “BAND.” No I in BAND. Another word is “TEAM.” No I. Another word is “CLAN.” No I. Another possibility is “GIRL” or “TRIBE.” TRIBE contains I. Removing I from TRIBE → TRBE (not a word). Another group word is “UNIT.” Contains I. Remove I → UNT.
3. Now “facing French street” suggests adding “rue” (French for street) or abbreviation “r.”
4. The clue likely involves inserting or appending the French street to the modified group word. ...
7. Another approach: “group I missed” could be “GANG” with I missed = GANG (no I). Or “GANG I” with I missed = GANG.
...
22. The clue might be an anagram indicated by “Crooked.”
23. The fodder might be “group I” + “rue.” Letters: G R
...
46. “Crooked” = ASKANT (6 letters) fits the definition and enumeration.
47. “Group I missed” = “A S K A N T”? No.
48. “Facing French street” = “rue” abbreviated “r.”
49. Possibly “group I missed” = “askant” is hidden or formed by removing I from “against” (group I missed).
50. “Against” (7 letters) remove I = “aganst” (6 letters) no.
51. The answer is ASKANT, meaning “crooked.” Therefore, the answer is ASKANT.

In the example above, the cryptic clue encodes a global deletion indicated by ‘missed’ and a local container indicated by ‘facing’. They should be unified in a single letter manipulation. However, in its reasoning trace, OS treats them as independent mechanisms and becomes tentative by trying different manipulation possibilities. As

inference progresses, the trace is less grounded in the clue and the mechanism identified. In the end, its derivation reflects no operations but a mere post-hoc rationalisation, like BS.

The limitation of OS in addressing cryptic clues with multiple wordplay signals is speculated to be related to the characteristic of the demonstration set in Chapter 5. While the curated reasoning demonstrations reflect valid and reconstructible inferences, most of them aim to solve single-mechanism clues. This thus induces a procedural transfer over isolated wordplay operations rather than over compositional inference strategies. Since most clues in the `test_data` related to 'anagram', 'alternation' and 'deletion' require coordination of local wordplay, the average IS scores for these classes decrease more saliently than the others.

Moreover, even if OS explicitly links letter transformation to cryptic conventions, it can be unfaithful in its explanation. For example, in (8), although OS names the device ('anagram') and mechanism correctly, its final derivation ('Output') cannot be traced back to the procedures it introduces (i.e., anagramming 'UTROOP').

(8) Clue: 'Remove disorderly scout troop missing northerner (6)'

- OS (IS = 0)

> 1. The definition is likely "Remove."
> 2. "Disorderly" is an anagram indicator.
> 3. "Scout troop" is the fodder, but "missing northerner" means remove a word or letters meaning "northerner" from "scout troop."
> 4. A common word for "northerner" is "Scot."
> 5. Remove "Scot" from "scout troop": SCOUT TROOP - SCOT = U T R O O P (letters left: U, T, R, O, O, P).
> 6. Anagram "UTROOP" (6 letters) indicated by "disorderly."
> 7. Anagram of UTROOP is OUTPUT.
> 8. OUTPUT means "Remove" (to output something is to remove it).

- Teacher-LLM

> 1. Definition: "Remove" at the end of the clue gives the solution's meaning.
> 2. "Northerner" in cryptic clues often clues SCOT.
> 3. The phrase "scout troop" supplies the fodder SCOUTTROOP.
> 4. "Missing northerner" instructs us to remove SCOT from SCOUTTROOP, leaving the letters U T R O O P.
> 5. "Disorderly" is an anagram indicator, so we anagram U T R O O P to get UPROOT.
> 6. UPROOT means "remove," matching the definition.

The example above suggests that the procedure transfer observed above in the positive examples remains shallow since OS does not really 'understand' what the procedure entails. In contrast, the corresponding teacher trace addresses this correctly.

Consequently, with curated reasoning demonstrations, the student-LLM seems to learn the operation of isolated wordplay. However, this remains in form and cannot be

generalised, particularly those clues showing a higher level of difficulty.

6.5 Structural Coherence

The last dimension of the reasoning quality focuses on structural coherence (SC). Here, a reasoning trace is evaluated in terms of both stepwise consistency and global organisation to construct the final answer, regardless of decomposition or inference correctness.

6.5.1 Quantitative Comparison

After having the judge-LLM score the reasoning traces generated by BS and OS on the `test_data`, the average SC scores per wordplay were automatically calculated (see Table 6.5.

Wordplay	**Nr.**	**BS-μ_{SC}**	**OS-μ_{SC}**	**+**	**-**	**=**
Anagram	62	**0.693**	0.548	5	**14**	43
Insertion	62	**0.581**	0.419	10	**20**	32
Container	62	**0.629**	0.468	8	**18**	36
Reversal	62	**0.613**	0.274	4	**25**	33
Hidden	62	**0.710**	0.532	8	19	**35**
Deletion	62	**0.548**	0.403	10	**19**	33
Homophone	62	0.548	**0.564**	**11**	10	41
Alternation	62	**0.677**	0.468	4	**17**	41

Table 6.5: Average SC Scores per Wordplay and Number of Score Changes per Wordplay

Compared to the other domains of reasoning quality analysed above, OS demonstrates a more consistent and significant decrease in the average SC for the majority of the classes, where ‘reversal’ stands out the most (25 out of 62 instances). This suggests a more significant negative effect of the curated demonstrations on the student-LLM to structure its reasoning trace to solve clues encoding ‘reversal’. However, there is a comparatively insignificant increase of 0.016 in the average SC score for ‘homophone’.

6.5.2 Qualitative Comparison

To investigate the behavioural patterns of OS that account for the overall decrease in SC, I manually inspected all the reasoning pairs with positive and negative differences.

Across all the 203 reasoning pairs extracted, OS presents its reasoning traces as ordered lists regardless of wordplay categories, which is only occasionally observed in the corresponding BS traces. This style shift is demonstrated in (9):

(9) Clue: '*Bear to join up* (4)'

- BS (SC = 0)

> The clue is "Bear to join up (4)".
> Step 1: Identify the definition.
> The definition is likely either "Bear" or "join up," as these are the two main components in the clue. Since the clue is short and the enumeration is (4), both "bear" and "join up" can be four-letter words.
> Step 2: Consider "join up" as the definition.
> "Join up" means to unite or connect, possibly "ally," "meld," "join," or "link." "Ally" is a good four-letter word meaning "join up" (to form an alliance).
> Step 3: Consider the wordplay.
> If the definition is "join up," then the wordplay should give "ally."
> Step 4: Analyze the wordplay "Bear to join up."Bear" can mean "to carry" or "to endure."
> Alternatively, "bear" can be represented by the abbreviation "AL" (from "Al" as a short form for "Albert" or "bear" is less likely to be abbreviated).
> Alternatively, "bear" can be "AL" from "bear" hidden inside or "bear" can be "AL" as an abbreviation is unlikely.
> ...
> Step 12: So "Bear to join up" might be a double definition clue: "Bear" (as in "ally" meaning to bear or support someone) and "to join up" (ally).
> Step 13: "Ally" means "to join up" and also can mean "bear" in the sense of supporting or carrying a burden.
> Step 14: Therefore, the answer is "ALLY."

- OS (SC = 1)

> 1. The definition is likely "Bear," meaning to tolerate or endure.
> 2. "to join up" suggests a word meaning to unite or connect.
> 3. A 4-letter word meaning both "bear" (as in tolerate) and "to join up" is "ABIDE."
> 4. ABIDE means "bear" (endure) and can metaphorically mean to stay or join up with a group.
> 5. ABIDE fits the enumeration and the clue.

While both reasoning traces reflect attempts to organise in a stepwise manner, BS can contain multiple sub-hypotheses under one main step. In contrast, OS numerises every step, sharing great resemblance to the presentation style in the demonstration set presented in Chapter 5. As a result, the second reasoning trace is more followable than the first, resulting in an increased score.

However, this imitation of ordered lists becomes quite superficial when OS struggles to settle on one hypothesis and generates extensively long reasoning traces, which can be seen in (10):

(10) Clue: '*Approximately the combined weight of Tory voters in 1983 returning Thatcher?* (8)'

- BS (SC = 1)

The clue is: ”Approximately the combined weight of Tory voters in 1983 returning Thatcher? (8)”
Step 1: Identify the definition.
Usually, the definition is at the start or the end of the clue. Here, ”Approximately” is likely the definition since the rest of the clue seems complex and ”Approximately” fits as a definition for an 8-letter word.
Step 2: Analyze the wordplay.
- ”the combined weight of Tory voters in 1983 returning Thatcher?”
- ”Tory voters in 1983” likely refers to ”Tories” or ”Tory” or ”votes” or something similar.
- ”returning” is a reversal indicator.
- ”Thatcher?” refers to Margaret Thatcher, often abbreviated as ”M” or ”Mrs” or ”Iron Lady” but more likely ”M” or ”Thatcher” as a whole.
Step 3: Consider ”combined weight” as a phrase meaning ”mass” or ”tonnage” or ”weight”.
Step 4: Try to break down ”the combined weight of Tory voters in 1983 returning Thatcher?”
- ”Tory voters in 1983” could be ”Tory” or ”Tories” or ”votes”.
- ”weight of Tory voters” can mean ”mass” or ”ton” or ”mass” or ”load”.
- ”returning Thatcher” suggests reversing ”Margaret” or ”Thatcher” or ”M”.
...
Step 11: ”the combined weight of Tory voters in 1983 returning Thatcher?”
- ”Tory voters in 1983” could be ”T” (Tory initial). - ”combined weight” could be ”mass”
. - ”returning Thatcher?” could be ”ET” (the last two letters of ”Thatcher” reversed ”rehtcaht” gives ”et” at the end).
Step 12: Put ”ESTIMATE” together: - ”ESTI” + ”MATE”
- ”Mate” can be ”Tory voters” (mates as voters or supporters).
- ”esti” is close to ”esti-” meaning approximate.
Step 13: Alternatively, ”estimate” means ”approximately” and is 8 letters.
Step 14: Conclusion: The answer is ESTIMATE.

- OS (SC = 0)

1. Definition likely at the start: “Approximately.”
2. “Tory voters in 1983” hints at a word or abbreviation for Conservatives or their voters; “Tory” often abbreviated as T.
3. “combined weight” suggests a word related to weight or measure.
...
9. Another synonym for approximately is “TOLERANT” (no), or “TOLERANT” is unrelated. 10. Another approach: “combined weight of Tory voters in 1983” might be “T” + “ORY” (voters) + “TON” (weight) = T + ORY + TON = TORYTON (not a word). 11. “returning Thatcher?” could be a reversal of “Maggie” or “Iron Lady” but no immediate fit.
...
22. Another idea: “Tory voters in 1983” = “TORY” + “83” (roman numerals LXXXIII) too complex.
23. Alternatively, “combined weight” = TON, “Tory voters in 1983 returning Thatcher?” indicates the word “MAGGIE” reversed = “EIGGAM.”
24. Could the answer be “ESTIMATE,” which fits “approximately” and is 8 letters?
25. Check if “estimate” can be derived: “T” (Tory) + “I” (Roman 1) + “MATE” (friend) + “S” (plural) + “E” (end letter). Not clear.
26. Another approach: “Tory voters in 1983 returning Thatcher?” = “TORIES” + “83” reversed + “Maggie” reversed? Too complex.
27. The answer is likely ESTIMATE, meaning “approximately.”
28. “Combined weight” = “TI” (weight units) + “MATE” (friend) + “S” (plural) + “E” (end). Possibly a loose wordplay.
29. Given the clue difficulty, the best fit is ESTIMATE.

Here, both BS and OS experimented with different hypothesised mechanisms and presented similarly to BS in (9). BS organises the reasoning trace into step-chunks,

where each chunk starts with an interpretation and proceeds with sub-hypotheses or trials. In contrast, OS simply enumerates every step without combining steps into meaningful chunks, making it harder to follow. Consequently, how OS presents here is quite illusory: the reasoning trace seems organised, but in essence the global organisation remains incoherent. This structural rigidity is observed in 61 of the 142 decreased instances extracted, suggesting that the structural imitation remains in form.

When it comes to local stepwise consistency, BS and OS do not show much difference, such as the traces in (10) above. Regardless of the presentation style, both reasoning traces are very unstable in wordplay analysis. When the first mechanism interpreted fails to construct a plausible answer, both models can abandon an analytical path without resolution and further revise the underlying interpretation. In the case of consistent failure, it can abruptly shift to post-hoc rationalisation by introducing and forcing validation of an answer.

However, this behaviour is not observed in the demonstration sets nor in the corresponding reference teacher traces, as shown below:

- Demonstration Clue: '*Mexican food thus, queen perhaps served up?* (5)'

> 1. Definition: "Mexican food" (5) suggests TACOS.
> 2. Wordplay: "thus" → SO.
> 3. "queen perhaps" → CAT (a "queen" is a breeding female cat in animal-breeding jargon).
> 4. Together "thus, queen perhaps" gives SO CAT.
> 5. "served up?" is a reversal indicator. Reverse SO CAT → TACOS.
> 6. TACOS matches the definition "Mexican food."

- Reference Teacher Trace: *Approximately the combined weight of Tory voters in 1983 returning Thatcher?* (8)'

> We read "Thatcher?" with a question mark as our (punny) definition – not Margaret Thatcher herself, but someone who "thatches" clothes: an iron-lady.
> The entire wordplay is then an &lit: the combined weight ("mass") of all the Tory voters who returned Mrs Thatcher in 1983 would, roughly speaking, be the Iron Lady. In other words, the mass (weight) of those who brought back Thatcher yields "Iron Lady," her well-known epithet.

As a result, the above observations suggest weak demonstrative signals in terms of structural coherence at the local level.

6.6 Summary

Taken together, compared to the standard zero-shot CoT, the effectiveness of the curated reasoning demonstrations generated by LLMs varies across wordplay types and

clues. Meanwhile, across all dimensions, the student-LLM remains incapable and even weaker in addressing certain wordplay categories.

Regarding behavioural changes, the student-LLM is more aligned with the teacher in style and form after the demonstration, particularly in explaining clue decomposition and presenting the reasoning process. In addition, the transfer of compositional inference seems to remain a superficial imitation of an isolated wordplay operation. These results and observations are further discussed in the following chapter.

Chapter 7

Discussion

As introduced earlier, the present thesis aims to investigate the effectiveness of augmenting curated reasoning traces generated by an LLM as few-shot demonstrations for another LLM. More specifically, how does it compare to the zero-shot CoT prompting (Sub-RQ1), and what reasoning behaviour is learnt (Sub-RQ2)?

This chapter aims to answer the proposed research questions by synthesising, interpreting, and reflecting on the results and observations obtained in the previous chapters.

7.1 Sub-RQ1

After the demonstration configuration, there is a general decrease across all dimensions (i.e., final accuracy, decomposition quality, inference soundness, and structural coherence). Across wordplay categories, there is a decrease in the average of 'reversal', 'deletion', and 'alternation' across all three dimensions of reasoning quality. However, it is hard to draw a universal conclusion concerning the performance difference after the curated few-shot demonstrations at the micro-level. Even within the same wordplay category, the effectiveness of the curated demonstrations can still vary between different clues.

7.1.1 Negative Utility in Task Non-Linearity

There is an overall performance degradation after the demonstration. This notion is proven by both quantitative and qualitative comparisons conducted in the previous chapter. Quantitatively, the aggregated score decreases across all dimensions, particularly for structural coherence. Qualitatively, the analysed reasoning pairs suggest that the student-LLM is likely to become less adaptive. It is also likely to become more

rigid to align with the curated demonstrations.

As a result, this degradation falsifies the hypothesis earlier raised on general performance gains. Moreover, while the optimisation process is more transparent and allows proactive human oversight, the claimed efficacy of the *Trace-of-Thought* framework in improving the student-LLM's problem-solving capabilities is supported to a limited extent (McDonald and Emami, 2024).

This unexpected performance degradation can be attributed to the way that a few-shot CoT correlates with task complexity. The evaluation benchmarks in McDonald and Emami (2024) are based on complex arithmetic reasoning, as demonstrated below (Cobbe et al., 2021) :

> *Question: Consider the function g(x) = 3. Find g(2).*
>
> Reasoning: Review the function definition: g(x) = 3; Identify the target: g(2) Substitute x with 2 in the function: g(2) = ? Acknowledge the function's constant value: ? = 3

Solutions to such questions are fundamentally linear and predictable, since they require a strict hierarchy of mathematical rules in a predetermined order. Meanwhile, every step builds directly on the previous premise to reach a single verifiable chain. Hence, providing demonstrations helps LLMs structure a valid mapping from input to output (Wei et al., 2022a).

In contrast, cryptic crossword clues demonstrate a significantly higher level of complexity. Although generally following fixed construction conditions and being predictable in the early segmentation (Efrat et al., 2021), intermediate inference, particularly wordplay disambiguation, can be quite non-linear and implicit. There is a need for simultaneous and dynamic shifts between surface-level components and abstract derivations.

Consequently, demonstrations may not necessarily provide a helpful reasoning template, even if they are generated by a capable solver and are well-curated. However, how task linearity and few-shot CoT effectiveness interact remains to be tested on a larger scale.

7.1.2 Clue-dependent effectiveness

Compared to the zero-shot CoT setting, I speculate that the impacts of curated few-shot demonstrations on the student-LLM are more instance-dependent than universal.

This notion is evident in the previous results and analysis. Within a wordplay category, reasoning pairs can be found showing both positive and negative score dif-

ferences. Moreover, intra-wordplay clue construction differences also play a role in affecting the demonstrative effectiveness in the 25 reasoning trace pairs analysed. Hence, quantitative changes at the average level are driven primarily by the relative proportion of improved and degraded cases, rather than by a uniformly positive or negative effect across all clues.

One possibility of such clue dependence is a mismatch in difficulty levels between the clues in the demonstration set and those in the `test_data`. In the small sample of reasoning pairs extracted to analyse effects, I observed that the post-demo student-LLM shows higher decomposition quality and greater persistence in wordplay operations when solving simpler clues. These clues show greater similarity to those in the demonstration set because they are explicit in components and mechanisms. As the difficulty level of a target clue exceeds that of the demonstration set, a shallow imitation becomes insufficient. As a result, the demonstration effect in such a scenario can be less positive, aligning with the claim of Zhou et al. (2022).

However, limitations during data processing in Chapter 3 should not be overlooked here. The creation of the `test_data` focused only on an equal distribution for all wordplay categories. However, other clue-contingent factors, including structural complexity and ambiguity, were not controlled as variables. This means that there are both relatively straightforward and highly ambiguous clues within a wordplay category, making it difficult to tell if the demonstrative effects are related to wordplay categories or clue difficulty. As a result, the second hypothesis (i.e., whether clue types involving letter transformation yield more performance gains under the proposed prompting paradigm) remains inconclusive, implying future investigation under a dataset with explicit difficulty inspection and stratification.

7.1.3 Internal Constraints

Despite the curated demonstrations, the student-LLM (i.e., GPT-4.1-mini) still displays a low-commitment reasoning style, which is consistently observed in the qualitative analysis in the previous chapter.

A likely reason for this phenomenon lies in its architectural constraints described in Section 2.1, as well as its designed purposes. GPT-4.1-mini, derived from GPT-4, is an autoregressive LLM that is trained for low latency and straightforward generation (OpenAI, 2026). Therefore, when experimenting with different speculations, it simply generates its speculations in a single pass without iterative refinements. Instead, its autoregressive nature forces it to continuously predict the next possible content based on its own prior mistakes, leading to repetition and incoherence. This aligns with the

limitations of few-shot learning presented by (Brown et al., 2020).

In comparison, the teacher-LLM, O4-mini, is trained with a different reasoning path. Instead of straightforward generation, it can recognise its mistakes, experiment with different strategies, and refine its thinking process internally (OpenAI, 2025). This helps explain its excellent performance during teacher selection in the diagnostic stage.

This interpretation implies that the internal reasoning pathway of LLMs is also an important factor to consider when deploying LLMs as in-context learners and teachers, beyond the focus on parameters suggested by (McDonald and Emami, 2024) and the task-specific capabilities in the present work.

Since Yao et al. (2023) demonstrates the efficacy of deploying LLMs as in-context teachers that exhibit similar internal pathways, I hypothesise that the demonstrative effect could be more salient and positive in GPT-5.4, which remains to be tested.

7.2 Sub-RQ2

Concerning behavioural changes after reasoning optimisation, the 25 reasoning pairs extracted for ‘homophone’ and ‘container’ hint that the student-LLM might have learned a concise explanation of the decomposition. Regarding inference, 12 of the 20 increased cases for ‘anagram’, ‘alternation’, ‘deletion’, and‘container’ signal that the student-LLM captured isolated wordplay operations from the demonstrations. Moreover, the 203 reasoning pairs compared consistently reveal the student-LLM’s high alignment with the demonstrations in the presentation format.

7.2.1 Structural Mimicry

The reasoning structuring process keeps providing a shallow imitation of the demonstrations without learning relevant strategies. When inspecting 61 of the 142 decreased cases for structural coherence, the post-demo student-LLM mechanically enumerates every step it takes without segmenting steps by main hypotheses, like its zero-shot version.

This strongly signals that the student-LLM remains weak in planning and organising its reasoning output even after the demonstration, despite alignment in form. Consequently, the curation demonstrations serve largely as an output format constraint, rather than inducing genuine structural reasoning trace generation. This thus resonates with the criticism of augmenting few-shot exemplars in CoT prompting by Cheng et al. (2025a).

7.2.2 Procedural but Non-Competent Learning

The student-LLM may also have captured the procedural signals of wordplay operations from the curated demonstrations. It carries out the captured procedures even towards the final attempt in the 12 examples analysed for 'anagram', 'alternation', 'deletion', and 'container'.

However, the comparison with the decreased cases (36) signals that these learned procedures do not extend to more complex cases, particularly when there are multiple fodders and mechanisms. In such cases, the student-LLM rigidly carries out the procedures in parallel without being able to coordinate wordplay interactions. In addition, it maintains a low-commitment behaviour regardless of the demonstrations.

Consequently, this contradicts the findings of McDonald and Emami (2024) that the student becomes more adaptive and capable of novel cases, nor does it fully validate the third hypothesis of procedural learning at the competent level.

A possible impediment to competence-level reasoning transfer beyond procedural can be associated with a persistent knowledge gap in handling cryptic clues. Sadallah et al. (2025) have revealed that cryptic-clue-solving remains challenging for LLMs because they can over-predict certain wordplay categories and cannot genuinely carry out an operation. Hence, to address this knowledge gap, tool augmentation can be a promising direction for knowledge injections (Ovadia et al., 2024).

Across all the sub-RQs, I reveal that the effectiveness of curating LLM-generated reasoning traces as demonstrations for another LLM is limited. Although procedural mimicry is salient, the improvement of reasoning competence to solve cryptic clues remains scarce. This leaves us with an important question to reflect on: with curated demonstrations, are LLMs genuinely learning to solve a task, or are we merely improving their adherence to our instructions? What has been revealed in the discussion above seems to align with the latter interpretation: the demonstrations are shaping how solutions are articulated rather than how they are derived.

7.3 Limitations & Future Work

There are still open questions regarding how the findings above generalise to other scenarios.

This primarily concerns the effects of data stratification on evaluation and analysis. Although the equal-allocation stratification adopted in the present work contributes to evaluation equality, intra-wordplay difficulty variance makes it hard to iso-

late wordplay categories and difficulty levels when attributing factors to performance differences. Future studies should take both categories and difficulty matching when stratifying data, and can test the correlation between difficulty and the demonstrative effects.

The efficacy of deploying LLMs as in-context teachers to curate few-shot demonstrations is also subject to model selection. The present work does not fully consider the effects of pathway similarity during model selection, nor does it experiment with different pairs of teacher-student models. Future research can consider deploying LLMs with similar pathways from different families. Moreover, the effects of manually crafted demonstrations on solving cryptic clues are not compared, which can be explored in future studies as well.

Further, although the present work contributes a fine-grained LLM-as-a-judge evaluation mechanism for cryptic clues, its wider applicability is still limited. In the context of this thesis, the current level of validity and reliability is sufficient for the purposes of this thesis. It deserves to be further worked on on a larger scale, helping to develop a robust, fine-grained evaluation metric for the domain.

Beyond these, future work should also examine the prompting framework in the present work in the domains that share a similar level of non-linearity to consolidate the speculated correlation between task complexity and framework utility. Moreover, future work can combine knowledge injection with the current prompting framework to see if there are more reasoning performance gains.

Chapter 8

Conclusion

In conclusion, this thesis shows that augmenting curated teacher-LLM traces in few-shot CoT prompting offers limited reasoning performance gains of the student-LLM in solving cryptic crossword clues. The curated demonstrations lead to marginal or negative effects on model performance in terms of final accuracy, decomposition quality, inference soundness, and structural coherence.

Across all eight wordplay categories, the student-LLM shows a broadly consistent shift in presenting reasoning chains. Aligning with the demonstrations, it presents the reasoning traces in ordered lists more frequently. However, this does not translate into a structurally coherent reasoning. It does not reliably segment steps into meaningful chunks when there are multiple hypotheses. Therefore, this suggests the structural learning remains largely superficial and serves more as output constraints.

In comparison, the demonstrative effects on decomposition quality and inference soundness are not uniform across wordplay categories and individual clues. Although some cases signal that the student-LLM captures procedural signals in the demonstrations to a certain degree, the behavioural changes do not generalise consistently. The post-demonstration student-LLM remains weak and tentative in decomposition and wordplay analysis, indicating limited transfers at the competence level.

Consequently, I speculate that the curated demonstrations play a more important role in shaping the reasoning process as instructed rather than improving capabilities at a deeper level to solve a clue.

However, whether the above findings are generalisable remains to be further explored. There is likely a complex interplay of task non-linearity, clue difficulty, model architectural constraints, internal pathway similarity, and knowledge gaps. The present work shows limitations in fully addressing all these perspectives, as well as exploring at a larger scale with multiple teacher-student pairs.

Despite the limitations described above, the current work contributes by addressing the gap concerning process-based supervision and optimisation in the field of cryptic-clue solving. Moreover, in using cryptic clues as a controlled environment, the present work helps expand the evaluation of the Trace-of-Thought framework by (McDonald and Emami, 2024) in linguistically complex tasks. More importantly, this work develops a fine-grained LLM-as-a-judge reasoning quality evaluation mechanism customised for cryptic-clue solving.

Future studies can proceed in investigating the proposed reasoning optimisation framework at a larger scale by focusing on its interaction with model selection, example curation, data stratification, and/or external knowledge injection. More broadly, there should be a broader investigation of reasoning optimisation in the scenario of complex and creative language use.

Finally, a more philosophical but fundamental question deserves our continuous exploration and reflection: *to what extent do reasoning optimisation methods bring LLMs closer to reasoning like human beings*, or *are they simply producing an advanced form of stochastic learning*?

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Appendix A

Lightweight Experiment (Exploratory)

A.1 Preliminary Dataset

Five fully annotated clues per wordplay were randomly selected from GHC, amounting to 40 instances.

A.2 Prompt and Configuration

A.2.1 Simplistic Modelling Prompt

To enforce models to generate intermediate reasoning steps for the clues sampled, the `ChainOfThought` module in DSPy (Khattab et al., 2023) was used. The following were added in the signature docstring and the field description as prompts:

Task description: *Solve the cryptic clue. Reason that it generally consists of two parts, the definition and the wordplay. The definition is directly related to the answer, like a synonym or hyponym, and the wordplay needs to be deciphered, usually with the help of an indicator that hints at what you need to do.*

InputField (Clue) description: *Cryptic clue followed by a bracket showing the number of letters needed in the answer.*

A.2.2 Model Configuration

Given the task complexity, the temperature for the models was set at 0.6 to retain reasoning stability while allowing for more creativity. However, reasoning models (i.e., DeepSeek-R1 and O4-mini) did not support temperature configuration. Hence, their

default temperature (1.0) was used. Other parameters, such as 'top-k' and 'reasoning_effort', were maintained as their default values.

A.2.3 Reasoning Outputs

The reasoning outputs can be viewed in the `data/diagnostic_exp` folder of the repository.

A.2.4 Preliminary Model Behaviour Exploration

Basic Metric

A basic binary metric was used for preliminary exploration, as presented in Table A.1.

Dimensions	Description
Factual Accuracy (0/1)	Whether the solver answer exactly matches the gold answer.
Faithfulness (0/1)	Whether all reasoning steps are grounded in explicit clue text or widely recognised cryptic crossword conventions. Invalid shortcuts and post-hoc/biased rationalisation are marked as unfaithful.
Global Correctness (0/1/2)	Global correctness of the reasoning using a 3-point Likert scale: **0**: key logical errors; **1**: partially correct with mistakes; **2**: logically correct overall.

Table A.1: Exploratory Experiment Basic Metric

Scores and Notes

Note: For O4-mini, global correctness was only qualitatively inspected.

LLM Preliminary Judge

As shown in Figure A.1, DSPy was used to prompt the model to generate justification before scoring. Meanwhile, the temperature was set at 0.2 to ensure adherence to the rubric.

Model	Acc.	Corr.	Faith.	Notes
GPT-5.4-mini	0.194	0.649	0.459	Much weight is given to definition and **searching for synonyms**; there is some incompleteness and **redundant thinking**, but it is better than 4o-mini.
GPT-4o-mini	0.054	0.649	0.432	Structured output; some good wordplay mapping; **hallucination**.
DeepSeek-R1	0.324	0.865	0.568	Structured output, explicit step segmentation in two reasoning traces, **admission of insufficiency but still gives a final answer**.
O4-mini	0.730	/	0.595	Strong elicitation/application of world knowledge; **compressed reasoning**; clue solving in parallel rather than entangling definition and wordplay.

Table A.2: Overview of Exploratory Stage Human Annotation Results

```
# Solver output
solver_reasoning: str = dspy.InputField(desc="Solver's full reasoning trace")
solver_answer: str = dspy.InputField(desc="Solver's proposed answer")

# Accuracy (answer-level)
factual_accuracy: int = dspy.OutputField(
    desc="0/1: Whether the solver answer exactly matches the gold answer"
)

# Faithfulness (grounding)
faithfulness: int = dspy.OutputField(
    desc=(
        "0/1: Whether all reasoning steps are grounded in explicit clue text "
        "or widely recognised cryptic crossword conventions. "
        "invalid shortcuts and post-hoc/biased rationalisation as unfaithful."
    )
)
faithfulness_rationale: str = dspy.OutputField(
    desc="Justification focusing on grounding, indicators, and abbreviations"
)

# Logical validity (independent of grounding)
validity: int = dspy.OutputField(
    desc=(
        "Global correctness of the reasoning using a 3-point Likert scale: "
        "0 = key logical errors; "
        "1 = partially correct with mistakes; "
        "2 = logically correct overall."
    )
)
validity_rationale: str = dspy.OutputField(
    desc="Explain logical correctness without repairing the reasoning"
)
```

Figure A.1: Exploratory Stage LLM Judge Prompt

Appendix B

Instrument Development (Diagnostic)

This appendix selectively presents the calibration configurations and results for the initial, middle, and final stages of metric development.

B.1 Initial Calibration

B.1.1 LLM Judge Configuration

- Model: GPT-4.1-mini
- Temperature: 0.3
- Additional task description per dimension: ”You are an expert cryptic crossword clue judge. You need to evaluate a solver’s reasoning trace. You are given the clue, the final answer, and useful information to assist evaluation. Focus on and rate the [XX] of the reasoning trace. DO NOT REPAIR REASONING.”

B.1.2 Metric

Dimensions	**What it measures**
Decomposition (0/1/2)	Whether model identifies/splits clue into definition, wordplay, indicator, fodder, and enumeration: • 0: Fails to identify/incorrectly identifies components. • 1: Partial identification with slight omissions. • 2: Correct and complete identification.
Inference (0/1/2)	Logic based on explicit spans and recognised cryptic indicator conventions: • 0: Invalid, major errors, or forced reasoning. • 1: Partially valid with minor errors. • 2: Logically valid and well-grounded throughout.
Coherence (0/1)	Consistent and structured reasoning trace: • 0: Disorganised or unstable (e.g., self-contradictions). • 1: Concise, coherent, and consistent flow.

B.1.3 Results

Below are the IAA scores calculated.:

- Decomposition κ : 0.18
- Inference κ : 0.31
- Coherence κ: 0.60

Apart from structural coherence, the agreement is at a weak level, following the standard interpretations.

Disagreed cases qualitative analysis

- In decomposition and inference, there are many 0-2/2-0 differences, implying that the metric might not be structurally stable. It was observed that there was dimension boundary leakage (i.e., checking inference soundness in decomposition).
- A solver might contain multiple attempts to solve the clue, necessitating the need to set constraints about which attempt to assess.
- One of the colleagues suggests the metric should be more rule-based and provide more evidence/examples for the model to look at.

Best traces selected (above 5)

- human-6: 1,2,35,40
- human-5: 5,28,30,31,36,37
- llm-6: 1,5,28,30,31,35
- llm-5: 37

Among the six best traces selected by the LLM, only two were also rated the highest by the human. Many of the best cases selected by the LLM did not receive a full score, according to the researcher. This means the selectional agreement is low and indicates limited alignment between LLM-based selection and human-rated best-trace identification, suggesting instability under the current metric version.

B.2 Middle Calibration

B.2.1 LLM Judge Configuration

- Models: GPT-4.1-mini and GPT-5.4-mini
- Temperature: 0.2
- Additional task description template per dimension (structured by DSPy): “Your input fields are:

1. ‘clue‘ (str): Cryptic clue to be solved;
2. ‘useful_information‘ (dict): Structured clue analysis (definition, indicator, wordplay type);
3. ‘answer‘ (str): Gold-standard answer
4. ‘solver_reasoning‘ (str): Full solver reasoning trace. May contain multiple attempts.

Your output fields are:

1. ‘reasoning‘ (str);
2. ‘[DIMENSION NAME]‘ (int): Final score. Must be exactly one of XXX. Determined strictly by the rubric in the task description.”

B.2.2 Metric

Dimensions	**What it measures**	**Scales**
Decomposition Quality (0/1/2)	Given the clue and the provided useful information, evaluate and justify whether the given reasoning correctly identifies and segments the following components: definition, wordplay, indicator (if present), and the required number of letters. Wordplay type classification belongs to inference, not decomposition. Only assess component identification and boundary detection, not the correctness of the underlying cryptic mechanism or final answer. No exact string matching is required; partial or paraphrased identification is acceptable as long as the component role is correctly inferred. If multiple reasoning attempts are present (evidenced by ‘However’, ‘Reassessing’ and ‘Alternatively’), evaluate the last stable segmentation that is explicitly used for the conclusion.	• **0**:Assigned when either: A) *Definition or wordplay* is missing, incorrectly identified, or conflated. OR B) The decomposition is not supported by the clue text (i.e., hallucinated or arbitrary structure). Example: invented definition or indicator spans • **1**: A) The reasoning correctly identifies both definition and wordplay as distinct components, even if imperfectly segmented AND B) At least one of the auxiliary components is incorrect or missing: indicator misidentified/missing OR letter count incorrect / omitted OR weak/ambiguous mapping of auxiliary structure • **2**: All components are extracted based on the clue text.
Inference Soundness (0/1/2)	*Evaluate and justify* whether the reasoning trace is mechanically valid and internally reconstructable under standard cryptic clue conventions. The given ’useful_information’ is used for reference only, since the routes to solving the wordplay can vary. Assume the decomposition is fixed. Do NOT evaluate the correctness of the definition and indicator identification! Focus on stepwise inference quality: GIVEN THE PREMISE, HOW IS THE INTERMEDIATE THINKING? Do NOT repair, reinterpret, or replace the reasoning with alternative solutions. If multiple reasoning attempts are present (evidenced by ‘However’, ‘Reassessing’ and ‘Alternatively’), evaluate the last stable segmentation that is explicitly used for the conclusion.	• **0**: Assigned when either: A) The clue contains invalid cryptic logic. Examples: answer-first reasoning without a valid derivation, incorrect letter counts, pure guessing or synonym substitution, considering active wordplay components misdirected/useless, etc. OR B) cannot be mechanically followed OR C) The reasoning is not clue-based. • **1**: Assigned when: A) the reasoning is directionally plausible; AND B) contains no more than two unclear, incorrect, or insufficiently supported steps. Examples: correct premise but incorrect inference, incomplete explanation of a transformation, partial but not fully justified letter manipulation • **2**: Assigned when the reasoning is fully mechanically valid and reconstructable. All transformations follow standard cryptic conventions and clearly support the final answer.
Structural Coherence (0/1)	Regardless of decomposition, reasoning correctness, whether the reasoning trace is consistent and structured.	• **0**: disorganised (e.g., early arrival of answers after extracting definitions) or unstable reasoning (e.g., self-contradictions, abrupt shifts, and redundant thinking). • **1**: coherent and consistent reasoning flow.

B.2.3 Results

The table below presents the IAA results in each model's evaluation run. While the two deployed models demonstrated approximately the same level of agreement with the researcher regarding clue decomposition and structure, GPT-5.4-mini's evaluation of inference soundness better aligned with that of the researcher's. As a result, GPT-5.4-mini will be used for subsequent calibrations.

Model	DQ_α	IS_α	SC_κ
GPT-4.1-mini	0.645	0.469	0.650
GPT-5.4-mini	0.593	0.791	0.699

Table B.1: Judge Candidate Comparison

Subsequently, since the agreement between the researcher and GPT-5.4-mini on decomposition quality was the lowest, the disagreed cases were manually inspected to further refine this dimension. Although there were more adjacent disagreements, two cases of 2-0 disagreements were still observed. Here is one of the corresponding justifications by GPT-5.4-mini:

- **Solver's reasoning**:

> The clue 'for ten years' suggests the answer is a term meaning a decade. The wordplay involves an anagram (indicated by 'abducted') of 'by alien' placed inside 'ET' (as in the movie), and 'I estimate' provides 'I' and 'EST' (abbreviation for estimate). Combining these elements: an anagram of 'by alien' (B, Y, A, L, I, E, N) gives 'belyin', inserting into 'ET' gives 'EbelyinT', and adding 'I' and 'EST' forms 'EbelyinTiest'. However, the correct anagram and combination leads to 'decennia', an 8-letter term for a ten-year period.

- **GPT-5.4-mini Justification**:

> The reasoning does identify the definition as 'for ten years,' which matches the clue's definition component. It also treats 'abducted by' as the indicator and attempts to parse the rest as wordplay. However, the segmentation is not supported by the clue text: it invents an anagram-based structure ('by alien' anagrammed, inserted into ET, plus I/estimate), whereas the provided analysis says the wordplay is a hidden answer. So the wordplay boundary/component identification is incorrect overall, even though the definition is correctly spotted. The required enumeration is correctly noted as 8 letters, but the actual decomposition does not align with the clue's indicated hidden-word structure. Since the definition is right but the wordplay is misidentified/conflated, this is not a fully correct decomposition.

It can be observed in the example that the judge also attempted to evaluate the correctness of the segmented wordplay chunk. However, since there was no gold annotation on wordplay material segmentation, it was hard to assess correctness. This needed to be refined. Also, the structural coherence descriptors were still vague.

B.3 Final Calibration

B.3.1 LLM Judge Configuration

- Model: GPT-5.4-mini
- Temperature: 0.2
- Additional task description template per dimension (structured by DSPy): same as above

B.3.2 Final Metric

See Table 4.2

B.3.3 Results

IAA Analysis Table B.2 presents the IAA scores for the final calibration round.

Model	DQ_α	IS_α	SC_κ
GPT-5.4-mini	0.738	0.800	0.684

Table B.2: Final Calibration IAA Scores

Disagreement Analysis The adjacent-score divergence (e.g., 0–1 and 1–2 discrepancies) was significant, implying ambiguity concerning boundary conditions. Manual inspections of the corresponding LLM-generated justifications further indicated that these disagreements were not random. Instead, the researcher and the LLM showed different strictness thresholds, particularly with regard to trace reconstructability and the acceptability of partially correct segmentations. Due to space constraints, the complete analysis was documented here.

Selection Overlap

- human-6: 1,2,5,35
- human-5: 28,30
- llm-6: 1,5,30,31,35
- llm-5: 2,28